PROMACH®

ORION®

Revision Date - May 21, 2020
Revision 3.1

Sentry HP & LP
STRETCHWRAPPER MANUAL

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Chapter 6: Electrical Prints and Mechanical Drawings

6. Electrical Prints and Mechanical Drawings

Electrical Prints

FILENAME	DWGDESC	DWGDESC2	DWGDESC3
PG1	COVER PAGE		
PG2	115VAC POWER DROP	AND DISTRIBUTION	100/200
PG3	115VAC VFD		300/400
PG4	START/STOP CIRCUIT	AND REMOTE START	500/600
PG5	X20 I/O POWER /	INPUTS	30100/30200
PG6	X20 I/O OUTPUT /	POWER SUPPLY	30300/30400
PG7	X20 RS485 COMM		30500/30600
PG8	CARRIAGE I/O BLOCK		50100/50200
PG9	QD CONNECTORS		
PG10	(CP1) PANEL AND BOM		
PG11	(CP3) ENCLOSURE	AND BOM	
PG12	MACHINE FIELD AND	CARRIAGE BOM	
PG13	CABLE CBL30502	(743145)	
PG14	CABLE CBL30502	(743145)	
PG15	OPTIONS PAGE		

WIRING COLOR AND SIZE
- 115VAC (RED) 14 AWG MINIMUM
- NEUTRAL (WHITE) 14 AWG MINIMUM
- GND (GREEN/YELLOW) 14 AWG MINIMUM
- +24VDC (BLUE) 18 AWG MINIMUM
- DC COMMON (WHITE/BLUE) 18 AWG MINIMUM
- CABLE COLORS AND GAUGE ARE LISTED IN PRINTS

DEVICE LOCATIONS
- (CP1) MAIN BACK PANEL
- (CP3) HMI ENCLOSURE
- (CARRIAGE)
- (FRAME)

NOTES FOR B&R PARTS
BRENTON TO PROVIDE AND INSTALL ALL THE FOLLOWING B&R PARTS
- HMI20115 B&R 4PPC.7070V1
- PLC30101 B&R X20BR9300 AND X20BM01
- PLC30201 B&R X20DIF371 AND X20BM11
- PLC30301 B&R X20DO9322 AND X20BM11
- PLC30401 B&R X20PS2100 AND X20BM01
- PLC30501 B&R X20CS1030 AND X20BM11

CHINA TO PROVIDE THE TERMINALS FOR THE B&R I/O
- PLC30101 B&R X20TB12
- PLC30201 B&R X20TB1F
- PLC30301 B&R X20TB12
- PLC30401 B&R X20TB12
- PLC30501 B&R X20TB12

CHINA TO PROVIDE THE COMM. CABLE INCLUDING CONNECTORS
- CBL30502
- CBL20123

NOTES FOR CARRIAGES
BRENTON CARRIAGE
- BRENTON WILL PROVIDE EVERYTHING ON CARRIAGE UP TO AND INCLUDING THE MALE 10 PIN QD FOR THE I/O AND THE 4 PIN QD FOR THE CARRIAGE MOTOR

NOTES
1. SILICONE BASE PRODUCTS SHALL NOT BE USED ANYWHERE IN THIS ASSEMBLY.
2. ALL FIELD CONTROL WIRE TO BE A MINIMUM 14 AWG 75C COPPER, UNLESS OTHERWISE NOTED.
3.
4. WHEN ORDERING REPLACEMENT PARTS, PLEASE SUPPLY THE JOB NUMBER AS SHOWN INSIDE THE CONTROL PANEL DOOR.
5. MANUFACTURES OF COMPONENTS USED IN THIS ASSEMBLY ARE SUBJECT TO CHANGE WITHOUT NOTICE.
6. THIS DOCUMENT IS THE PROPERTY OF THE END USER, PLEASE DO NOT REMOVE FROM THE ELECTRICAL CONTROL ENCLOSURE.
7. ALL WIRES CONNECTED TO THE B&R X20 IO MUST HAVE FERRULS.

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NO.	DATE	REVISION	BY

DWG TITLE
FLEX (STANDARD)
B&R SENTRY LP/HP

COVER PAGE

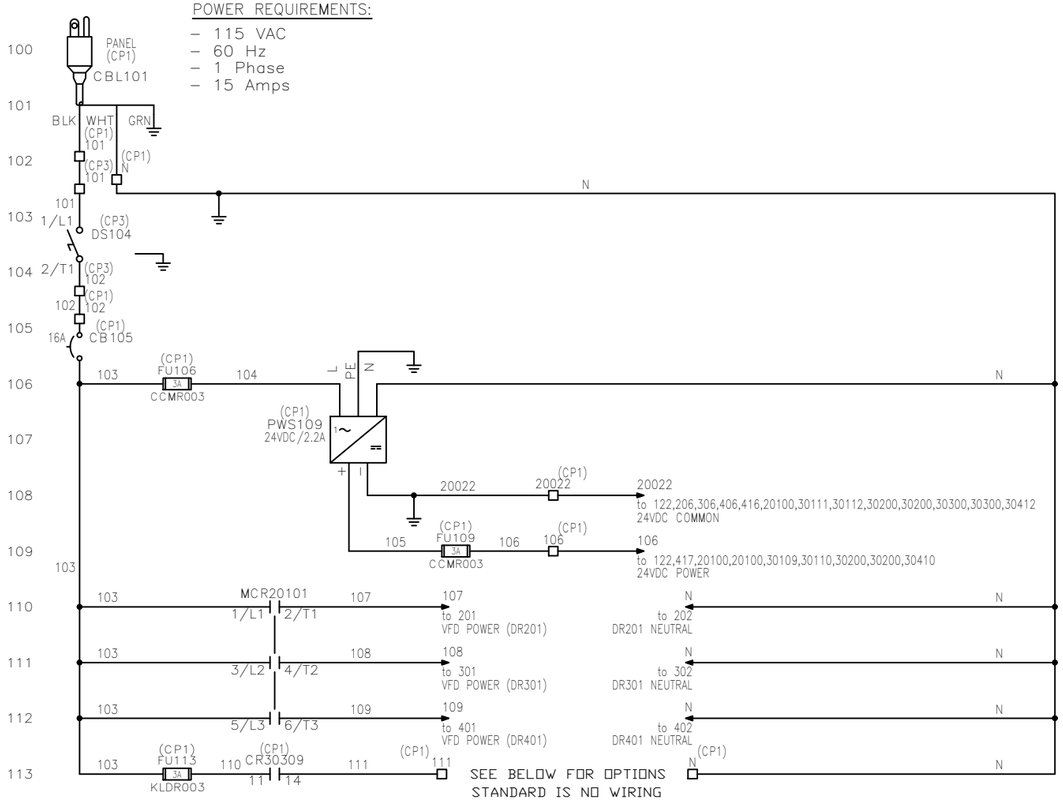
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SCALE: NO SCALE	DATE: 3/17/2018

DWG NO
B&R SENTRY

SHEET NO
1 OF 15

CAD DWG FILE I:\PROJECT_Flex Machine Orders\V6092 NEW SENTRY - Scale, ramp, manual\Electrical\schematics\PG1.dwg

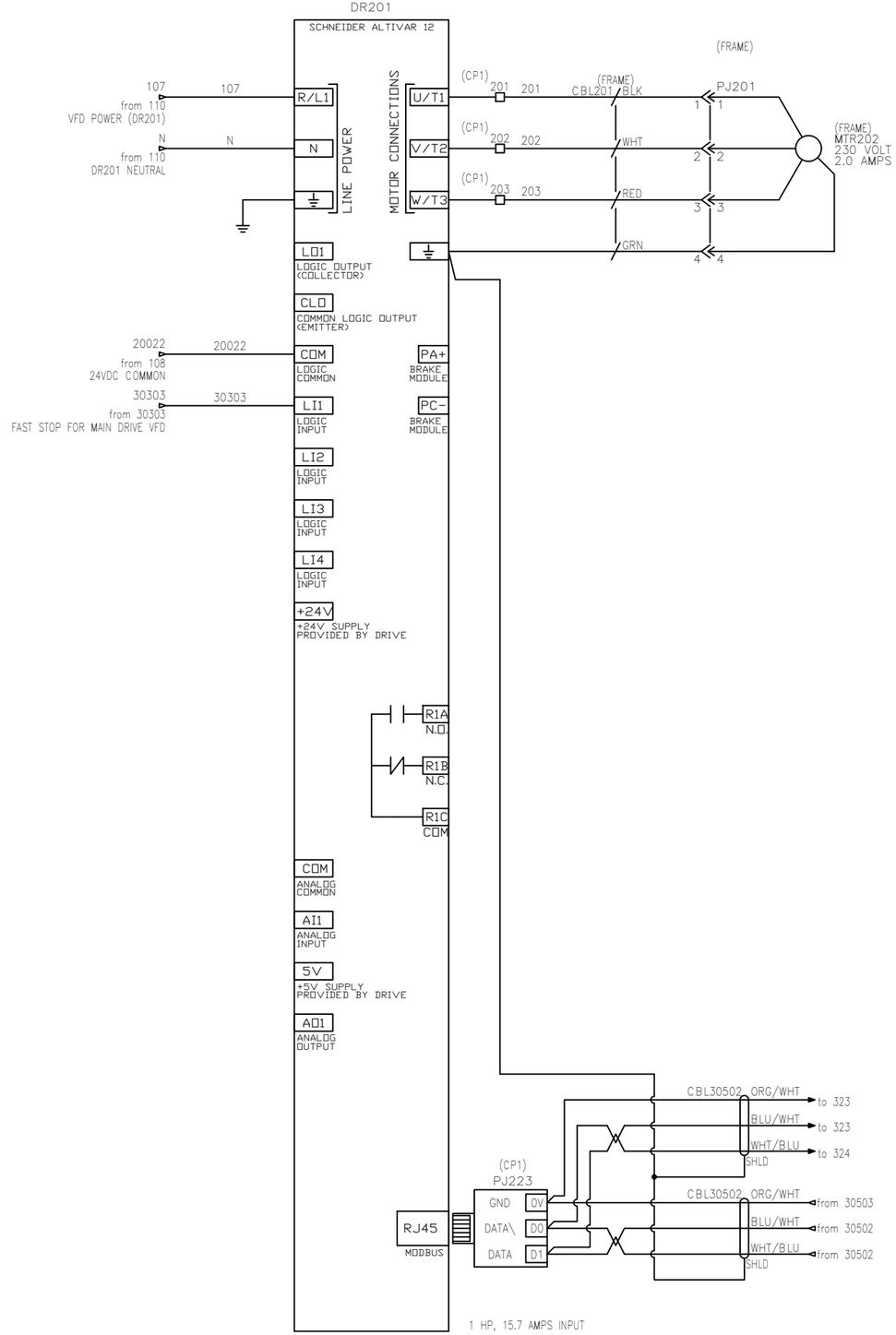
POWER REQUIREMENTS:
- 115 VAC
- 60 Hz
- 1 Phase
- 15 Amps



INSTA CUT OPTION
120 VOLT AC
ORION CARRIAGE ONLY

MAIN DRIVE

(CP1)
DR201
SCHNEIDER ALTIVAR 12



1 HP, 15.7 AMPS INPUT

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FLEX (STANDARD) B&R SENTRY LP/HP

115VAC POWER DROP AND DISTRIBUTION

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2 OF 15

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FILM CARRIAGE UP/DOWN

(CP1)
DR301

SCHNEIDER ALTIVAR 12

108 108 from 111 VFD POWER (DR301)
N N from 111 DR301 NEUTRAL
R/L1
N
LINE POWER

MOTOR CONNECTIONS
U/T1 — (CP1) 301 301 — (FRAME) CBL301 BLK
V/T2 — (CP1) 302 302 — WHT
W/T3 — (CP1) 303 303 — RED
GRN
(FRAME) MTR302 230 VOLT 2.0 AMPS

LO1 LOGIC OUTPUT (COLLECTOR)
CLO COMMON LOGIC OUTPUT (EMITTER)
20022 20022 from 108 24VDC COMMON
COM LOGIC COMMON
PA+ BRAKE MODULE
30301 30301 from 30301
LI1 LOGIC INPUT
PC- BRAKE MODULE
LI2 LOGIC INPUT
LI3 LOGIC INPUT
LI4 LOGIC INPUT
+24V +24V SUPPLY PROVIDED BY DRIVE

R1A N.O.
R1B N.C.
R1C COM

COM ANALOG COMMON
AI1 ANALOG INPUT
5V +5V SUPPLY PROVIDED BY DRIVE
AO1 ANALOG OUTPUT

(CP1) PJ323
RJ45 MODBUS
GND 0V
DATA\ D0
DATA D1
CBL30502 ORG/WHT — to 423
BLU/WHT — to 423
WHT/BLU — to 424
SHLD
CBL30502 ORG/WHT — from 221
BLU/WHT — from 221
WHT/BLU — from 222
SHLD

1/2 HP, 9.3 AMPS INPUT

MULTISTRETCH

(CP1)
DR401

SCHNEIDER ALTIVAR 12

109 109 from 112 VFD POWER (DR401)
N N from 112 DR401 NEUTRAL
R/L1
N
LINE POWER

MOTOR CONNECTIONS
U/T1 — (CP1) 401 401 — (FRAME) CBL401 BLK 1 — PJ401 1 1
V/T2 — (CP1) 402 402 — BLK 2 — 2 2
W/T3 — (CP1) 403 403 — BLK 3 — 3 3
GRN — 4 4
(FRAME)
(CARRIAGE) MTR402 230 VOLT 2.0 AMPS

LO1 LOGIC OUTPUT (COLLECTOR)
CLO COMMON LOGIC OUTPUT (EMITTER)
20022 20022 from 108 24VDC COMMON
COM LOGIC COMMON
PA+ BRAKE MODULE
30306 30306 from 30306 MULTISTRETCH VFD RUN
LI1 LOGIC INPUT
PC- BRAKE MODULE
30305 30305 from 30305 MULTISTRETCH SPEED REFERENCE
LI2 LOGIC INPUT
30307 30307 from 30307 MULTISTRETCH SPEED REFERENCE
LI3 LOGIC INPUT
LI4 LOGIC INPUT
+24V +24V SUPPLY PROVIDED BY DRIVE

R1A N.O.
R1B N.C.
R1C COM

20022 20022 from 108 24VDC COMMON
106 106 from 109 24VDC POWER
PRS417 BLU BRN BLK
PANEL (CP1) 417 417
(CARRIAGE) FILM DANCER BAR ANALOG PROX (ORION CARRIAGE ONLY)

COM ANALOG COMMON
AI1 ANALOG INPUT
5V +5V SUPPLY PROVIDED BY DRIVE
AO1 ANALOG OUTPUT

(CP1) R420 120 OHMS
(CP1) PJ423
RJ45 MODBUS
GND 0V
DATA\ D0
DATA D1
CBL30502 ORG/WHT — from 321
BLU/WHT — from 321
WHT/BLU — from 322
SHLD

1/2 HP, 9.3 AMPS INPUT

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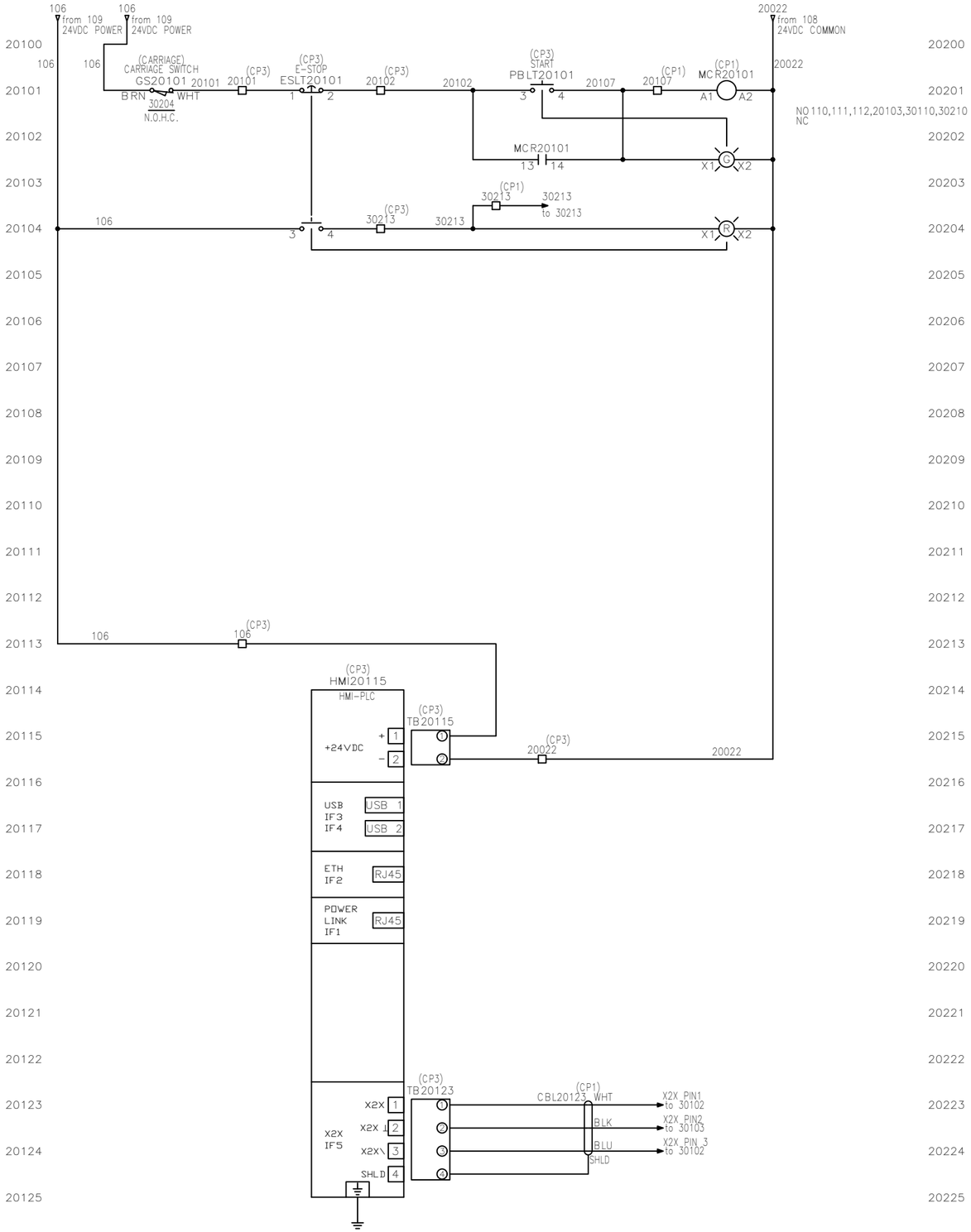
115VAC VFD

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FLEX (STANDARD) B&R SENTRY LP/HP

START/STOP CIRCUIT AND REMOTE START

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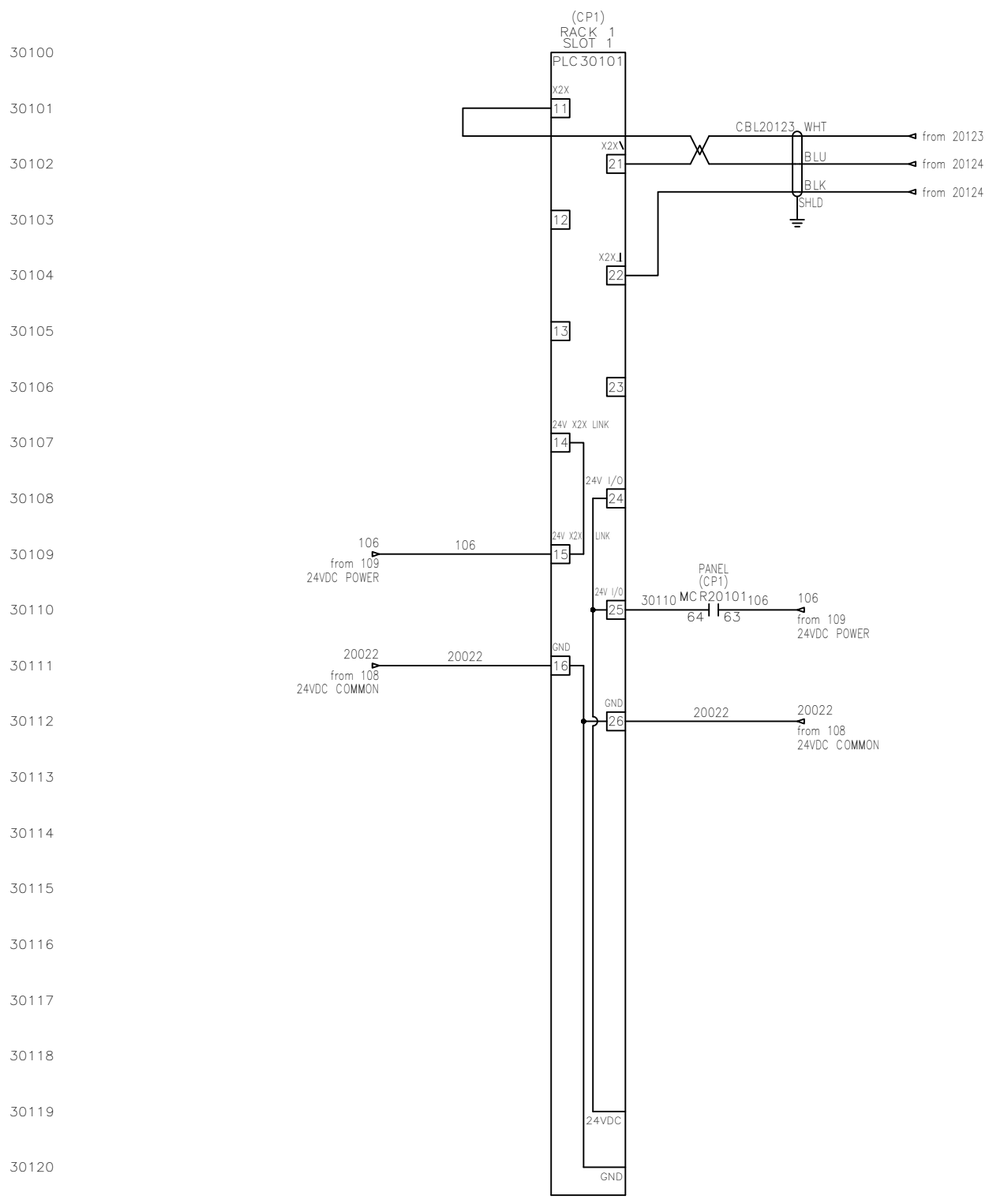
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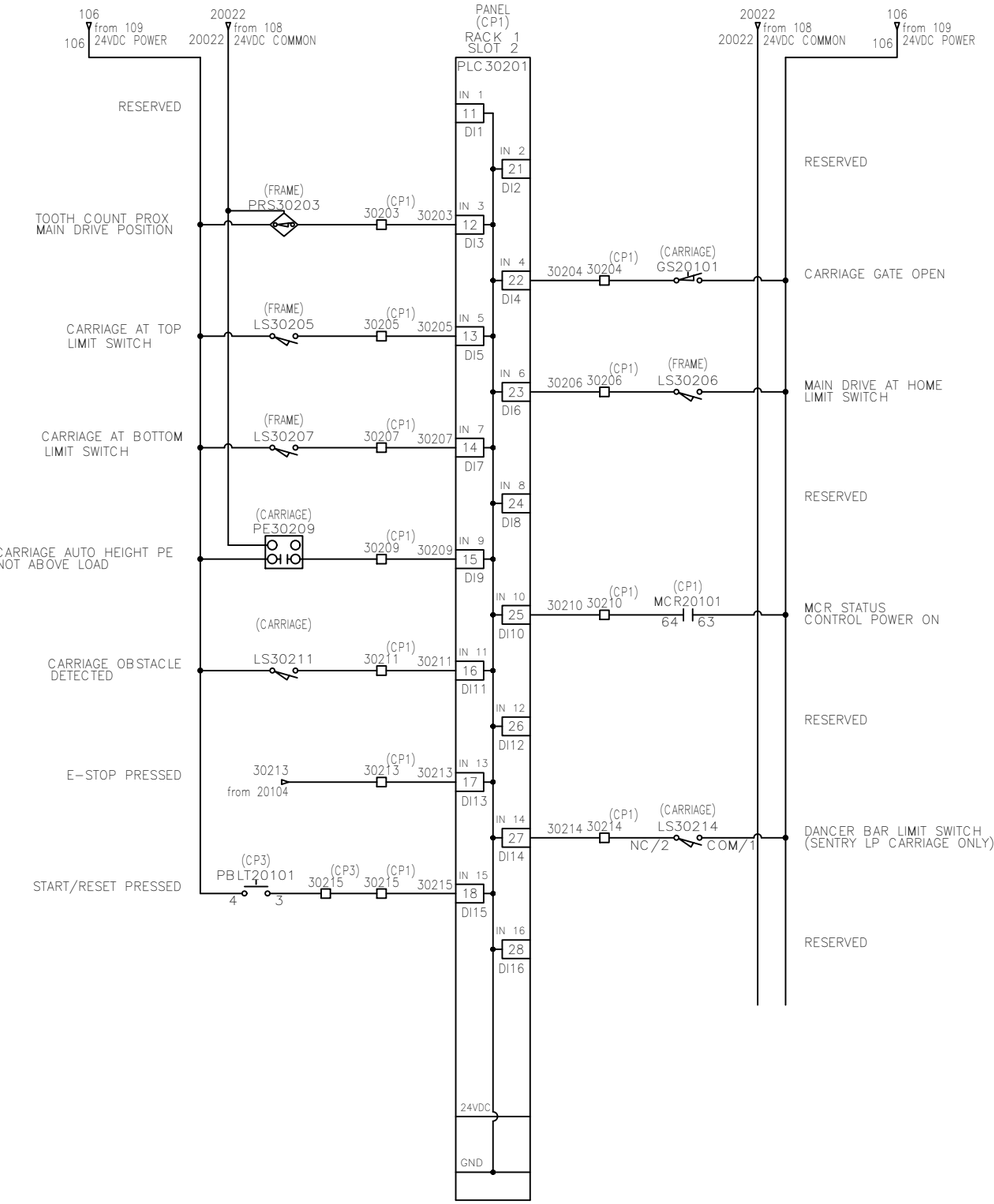
B&R SENTRY

SHEET NO

4 OF 15

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ALL WIRES CONNECTED TO THE B&R X20 IO MUST HAVE FERRULS.

ALL WIRES CONNECTED TO THE B&R X20 IO MUST HAVE FERRULS.

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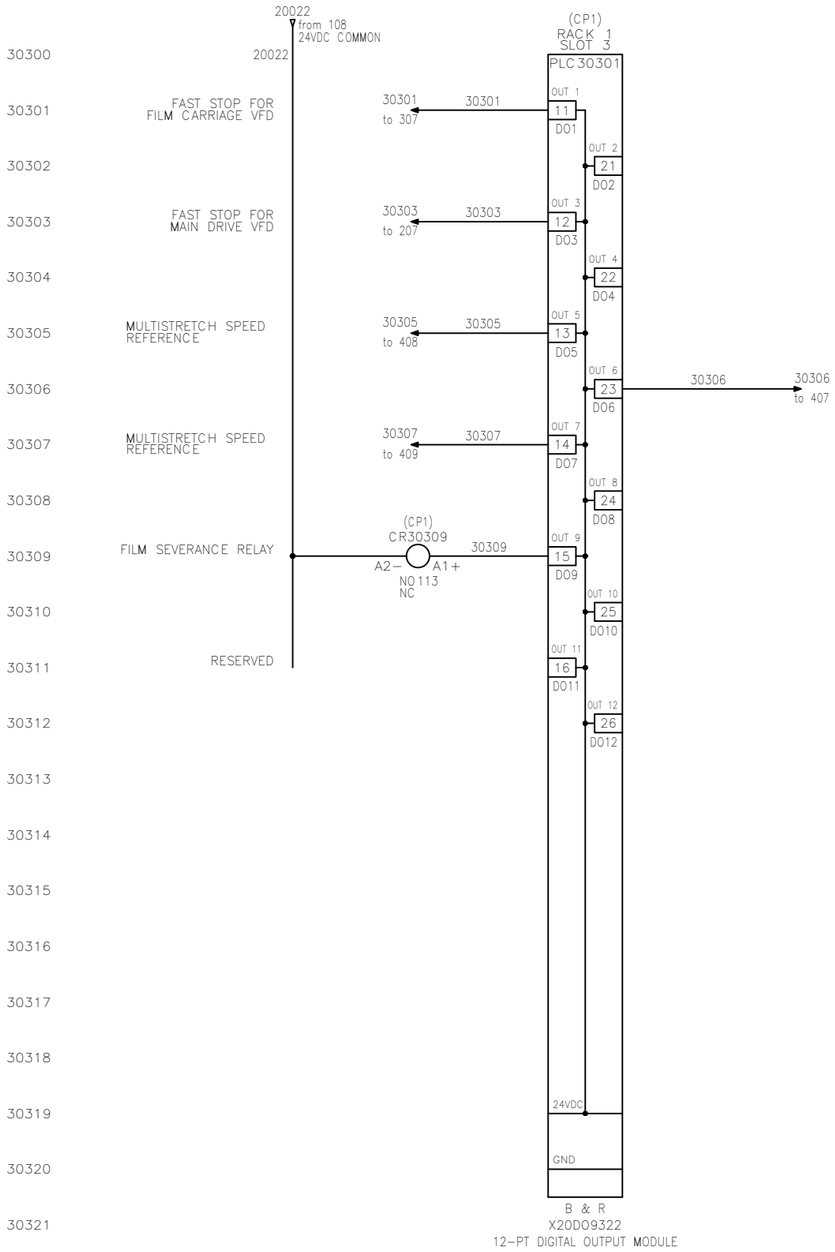
X20 I/O POWER / INPUTS

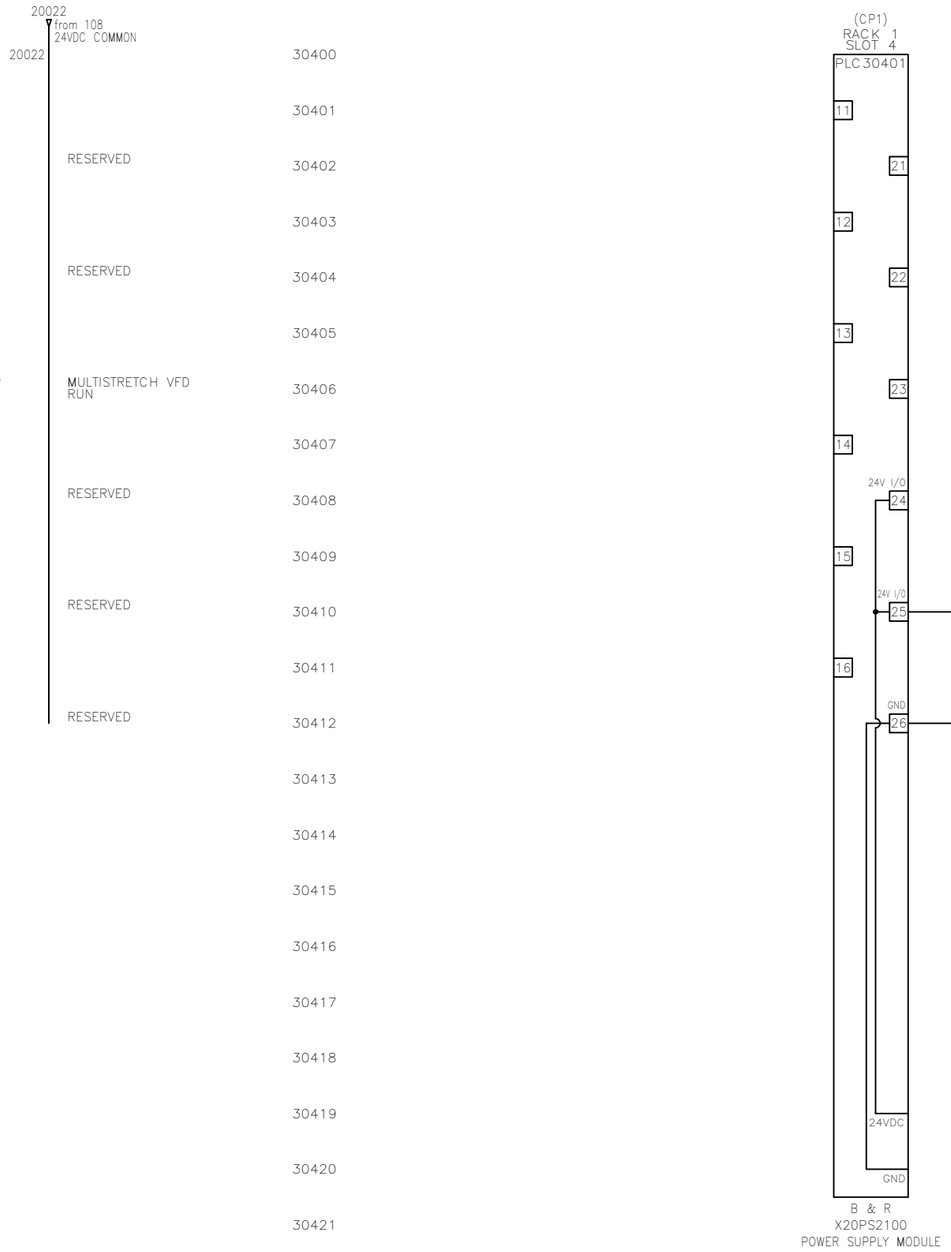
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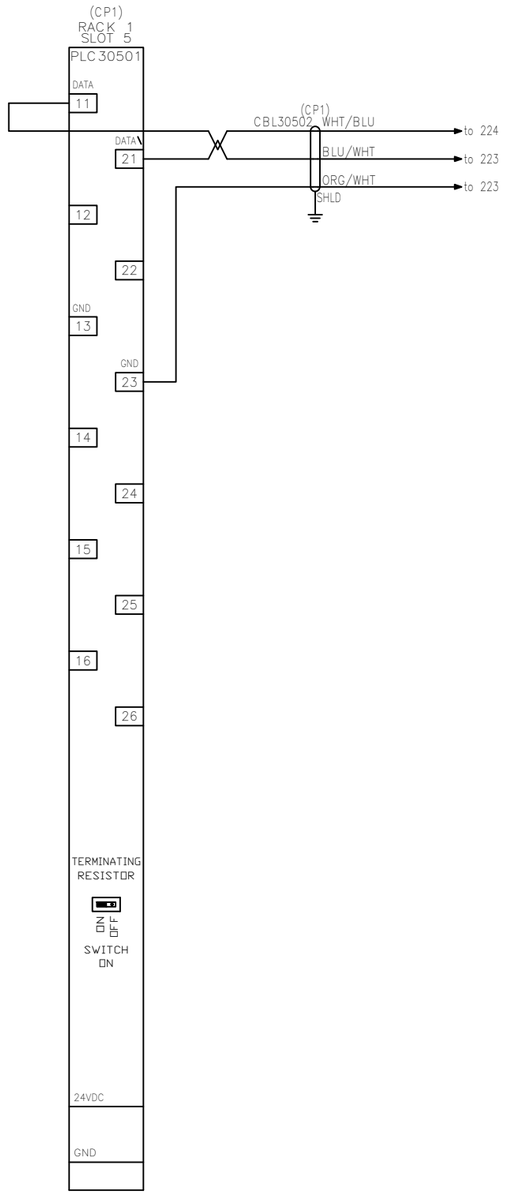
FLEX (STANDARD)
B&R SENTRY LP
X20 I/O OUTPUT /
POWER SUPPLY

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CAD DWG FILE I:\PROJECTS_Flex Machine Orders\V6092 NEW SENTRY - Scale, ramp, manual\Electrical\schematics\PG6.dwg



30500 30501 30502 30503 30504 30505 30506 30507 30508 30509 30510 30511 30512 30513 30514 30515 30516 30517 30518 30519 30520 30521 30522 30523 30524 30525

30600 30601 30602 30603 30604 30605 30606 30607 30608 30609 30610 30611 30612 30613 30614 30615 30616 30617 30618 30619 30620 30621 30622 30623 30624 30625

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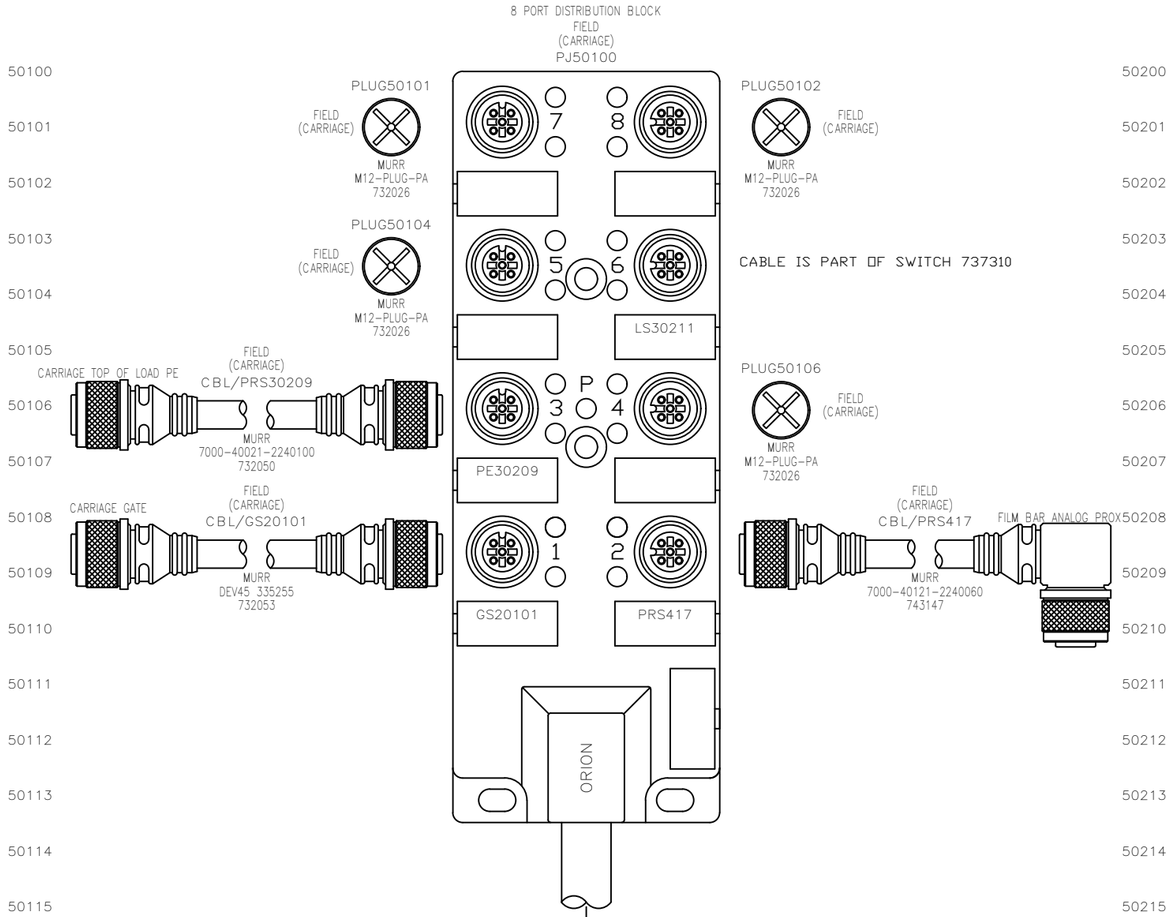
FLEX (STANDARD)
B&R SENTRY LP/HP
X20 RS485 COMM

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B&R SENTRY

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WIRE COLOR	PORT#	PIN#	DEVICE TAG	WIRE#	DESCRIPTION	I/O ADDRESS	PIN# ON QD
WHITE	1	4	GS20101	30204	CARRIAGE GATE OPEN		3
GRY/PNK	1	2	GS20101	20101	CARRIAGE GATE CLOSED		4
GREEN	2	4					
RED/BLU	2	2	PRS417	417	ANALOG PROX		9
YELLOW	3	4	PE30209	30209	CARRIAGE AUTO HEIGHT		5
WHT/GRN	3	2					
GRAY	4	4					
BRN/GRN	4	2					
PINK	5	4					
WHT/YEL	5	2					
RED	6	4	LS30211	30211	CARRIAGE OBSTACLE DETECT		6
YEL/BRN	6	2					
BLACK	7	4					
WHT/GRY	7	2					
VIOLET	8	4					
GRY/BRN	8	2					
BROWN	1 – 8	1		106	24VDC +		1
BLUE	1 – 8	3		20022	0VDC –		2
GRN/YEL	1 – 8	5		GND	GROUND		10

PIN 7 & 8 ON QD IS NOT USED

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DWG TITLE

FLEX (STANDARD)
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CARRIAGE I/O BLOCK

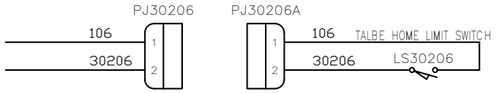
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B&R SENTRY

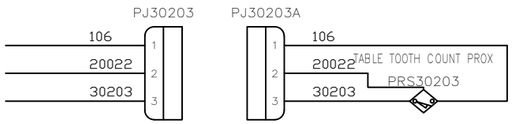
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8 OF 15

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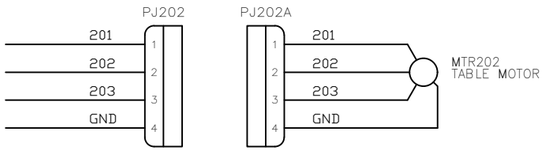
QD FOR TABLE HOME LIMIT SWITCH (LS30206)



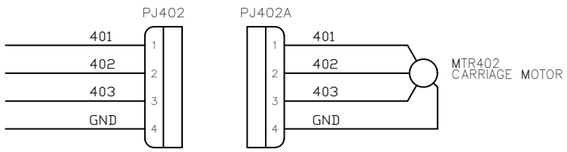
QD FOR TABLE TOOTH COUNT PROX (PRS30203)



QD FOR TABLE MOTOR (MTR202)



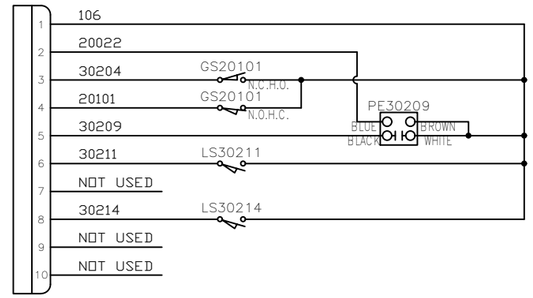
QD FOR CARRIAGE MOTOR (MTR402)



CABLE IN TOWER GOING TO CARRIAGE WITH QD

		PJ-CARRIAGE	
+24 VDC	BLACK	106	1
VDC COMM	BROWN	20022	2
CARRIGE GATE OPEN	BLUE	30204	3
CARRIAGE GATE CLOSED	WHITE	20101	4
TOP OF LOAD PE	PINK	30209	5
CARRIAGE OBSTACLE DETECT	GRAY	30211	6
FILM CUT	VIOLET	122	7
DANCER BAR LIMIT SWITCH	GREEN	30214	8
DANCER BAR ANALOG PROX	RED	417	9
GND	YELLOW	GND	10

QD ON CARRIAGE FOR STANDARD SENTRY CARRIAGE



QD ON CARRIAGE FOR SENTRY PLUS CARRIAGE

PJ-CARRIGE-B

1	106	BROWN
2	20022	BLUE
3	30204	WHITE
4	20101	GRY/PNK
5	30209	YELLOW
6	30211	RED
7	122	BLACK
8	NOT USED	
9	417	RED/BLU
10	GND	GRN/YEL

WIRED TO MURR QD BLOCK
SEE PAGE 8 FOR LAYOUT

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DWG TITLE

FLEX (STANDARD)
B&R SENTRY LP/HP

QD CONNECTORS

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JOB NO	DRAWN BY
SCALE: NO SCALE	DATE: 3/17/2018

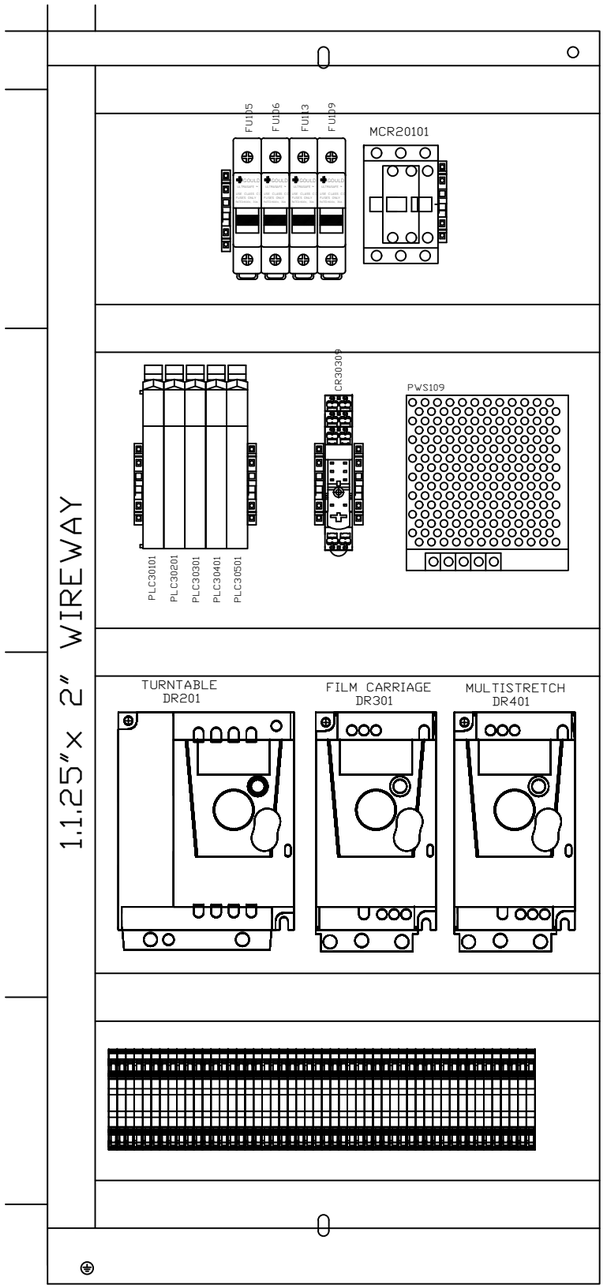
DWG NO

B&R SENTRY

SHEET NO

9 OF 15

CAD DWG FILE: I:\PROJECT_Flex Machine Orders\V6092 NEW SENTRY - Scale, ramp, manual\Electrical\schematics\P09.dwg



TERMINAL LIST
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20022

TAGS	QTY	SUB	ASSYCODE	MFG	CATALOG	DESCRIPTION	DESC2
CB105	1		750569	CHINT	NXB-63Z / C16		
CR30309	1	*1	75576	SCHNEIDER	RXM2LB1BD		
		*1	750577	TEND	PYF-08N		
DR301 DR401	2		732691	SCHNEIDER	ATV12H037F1	VARIABLE SPEED DRIVE, .5 HP, 0.37 kW, 120V, 1 PH, ATV-12 .5 HP 0.7KW,1HP	
DR201	1		732692	SCHNEIDER	ATV12H075F1	VARIABLE SPEED DRIVE, 1 HP, 0.75 kW, 120V, 1 PH, ATV-12 1 HP 0.7KW,1HP	
FU113	1	*1	750570	JINSHAN	RT18-32		
		*1	731039	LITTELFUSE	KLDR003	FUSE, CLASS CC, 3 AMP, TIME DELAY TIME DELAY 600VAC 3AMPS	
FU106 FU109	2	*1	750570	JINSHAN	RT18-32		
		*1	730995	LITTELFUSE	CCMR003	DUAL ELEMENT FUSE - CLASS CC, 3 AMP FAST ACTING, TIME DELAY 600VAC 3AMPS MOTOR CIRCUIT PROTECTION	
CBL101	1		738880	HARTFIEL	9-1051193		OR EQUIVALENT
PJ223 PJ323 PJ423	3		746197	B & R	8I0XC002.100.1		SUPPLIED BY HANGZHOU YOUNSUN SUPPLIED BY HANGZHOU YOUNSUN SUPPLIED BY HANGZHOU YOUNSUN
MCR20101	1	*1	732928	SCHNEIDER	LC1D09		OR EQUIVALENT
		*1	012635	SCHNEIDER	LADN20		
PLC30101	1	*1	731535	B & R	X20BR9300	X20 BUS RECEIVER, X2X LINK BUS RECEIVER, 24VDC X20 BUS RECEIVER	SLOT 1
		*1	742087	B & R	X20BM01		
		*1	742088	B & R	X20TB12		
PLC30501	1	*1	738900	B & R	X20CS1030	X20 INTERFACE MODULE, 1-PORT RS422/485, 24VDC, 115.2kbits X20 INTERFACE MODULE	SLOT 5
		*1	742083	B & R	X20BM11	PLC B&R BUS MODULE X20 TEREMINAL BLOCK	
		*1	742088	B & R	X20TB12		
PLC30201	1	*1	742084	B & R	X20DIF371	16 POINT X20 INPUT MODULE X20 DISCRETE INPUT	SLOT 2
		*1	742083	B & R	X20BM11	PLC B&R BUS MODULE X20 TEREMINAL BLOCK	
		*1	742082	B & R	X20TB1F	16 PIN TERMINAL BLOCK X20 TEREMINAL BLOCK	
PLC30301	1	*1	742996	B & R	X20DO9322	X20 OUTPUT MODULE, 12-PT DIGITAL SOURCE, 24VDC X20 DIGITAL OUTPUT	SLOT 3
		*1	742083	B & R	X20BM11	PLC B&R BUS MODULE X20 TEREMINAL BLOCK	
		*1	742088	B & R	X20TB12		
PLC30401	1	*1	738899	B & R	X20PS2100	X20 POWER SUPPLY, FOR INTERNAL I/O, 24VDC, X20 POWER SUPPLY Required line fuse: Max. 10 A, slow-blow	SLOT 4
		*1	742087	B & R	X20BM01		
		*1	742088	B & R	X20TB12		
PWS109	1		750572	SCHNEIDER	ABL2REM24020K		
R420	1		ANY	120 OHM	0.25 W		SUPPLIED BY BRENTON
	17		742993	PHOENIX	3209578		
	1		742994	PHOENIX	3209594		
CBL20123	2		743144	HARTFIEL	THC50967		SUPPLIED BY HANGZHOU YOUNSUN
CBL30502	6		743142	HARTFIEL	THC80290		SUPPLIED BY BRENTON

CBL20123 ASSY. INCLUDES
TB20123
AND CABLE

CBL30502 ASSY. INCULDES
PJ223
PJ323
PJ423
R420
AND CABLE

NOTE:
BACK PANEL INSTALLED
INSIDE TOWER UNIT

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DWG TITLE
FLEX (STANDARD)
B&R SENTRY LP/HP

(CP1) PANEL AND BOM

ENGINEER	CHECKED BY
JOB NO.	DRAWN BY
SCALE: NO SCALE	DATE: 3/17/2018

DWG NO.
B&R SENTRY

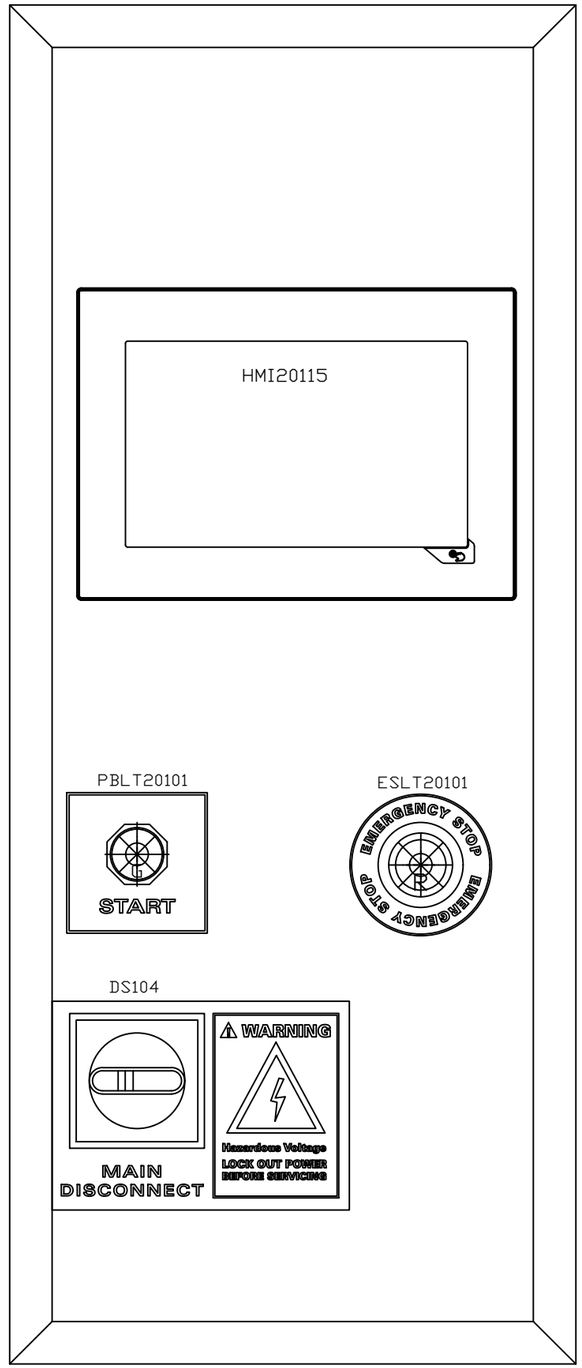
SHEET NO.
10 OF 15

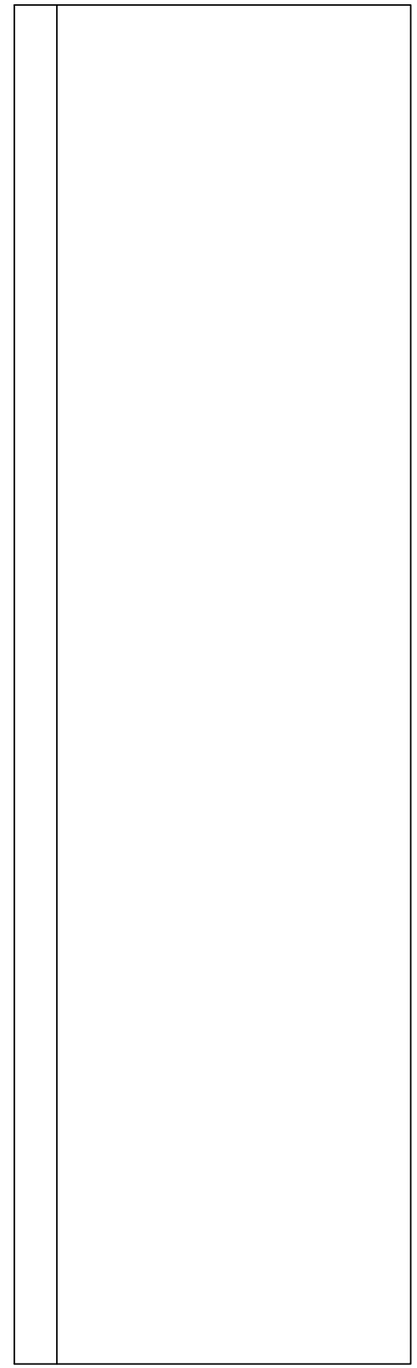
CAD DWG FILE I:\PROJECTS_Flex Machine Orders\V6092 NEW SENTRY - Scale, ramp, manual\Electrical\schematics\PG10.dwg

TAGS	QTY	SUB	ASSYCODE	MFG	CATALOG	DESCRIPTION
DS104	1		750571	Mei Wa	LW26GS-20	
TB20123	1		738894	B & R	□TB5104.2110-01	
TB20115	1		738883	B & R	□TB6102.2110-01	
HMI20115	1		743000	B & R	4PPC70.0702-20B	
PBLT20101	1	×1	017697	SCHNEIDER	ZB4BW333	
		×1	016210	SCHNEIDER	ZB4BZ009	
		×1	749762	SCHNEIDER	ZBVB35	LIGHT BLOCK - Ø22 - INTEGRAL LED GREEN - 24V
		×2	750291	SCHNEIDER	ZBE1015	CONTROL BUTTON CONTACT BLOCK - ZBE Ø22 - 1NO
ESLT20101	1	×1	019116	SCHNEIDER	ZB4BW643	
		×1	016210	SCHNEIDER	ZB4BZ009	
		×1	749763	SCHNEIDER	ZBVB45	
		×1	750291	SCHNEIDER	ZBE1015	CONTROL BUTTON CONTACT BLOCK - ZBE Ø22 - 1NO
		×1	749889	SCHNEIDER	ZBE1025	CONTROL BUTTON CONTACT BLOCK - ZBE Ø22 - 1NC
	2		742993	PHOENIX	3209578	

HMI ENCLOSURE AND ARTWORK TO BE PROVIDED WITH MACHINE.

TERMINAL LIST
106
106
106
106
20022
20022
20022
2011
20102
20102
20107
20107
30213
30213
101
102
GND





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NO.	DATE	REVISION	BY

DWG TITLE
FLEX (STANDARD) B&R SENTRY LP/HP

(CP3) ENCLOSURE AND BOM

ENGINEER	CHECKED BY
JOB NO	DRAWN BY
SCALE NO SCALE	DATE 3/17/2018

DWG NO
B&R SENTRY

SHEET NO
11 OF 15

CAD DWG FILE I:\PROJECT_Flex Machine Orders\V6092 NEW SENTRY - Scale, ramp, manual\Electrical\schematics\PG11.dwg

FIELD FRAME BOM

TAGS	QTY	SUB	ASSYCODE	MFG	CATALOG	DESCRIPTION	DESC2
PJ201	1		TBD	TBD	TBD		
LS30205 LS30206 LS30207	3		746319	OMRON	D4V-8108SZ-N		
MTR202 MTR302	2		SEE	MECHANICAL	BOM		2.0 AMPS 2.0 AMPS
PRS30203	1		750573	S&D	SN12-04BPA-Y		
CBL401	1	*0	739676	LUTZE	111371		
CBL201 CBL301	2	*0	730915	SOUTH WIRE	16/4 SLEOOW		
CBL115	1	*0	TBD	TBD	TBD		

FIELD CARRIAGE BOM

TAGS	QTY	SUB	ASSYCODE	MFG	CATALOG	DESCRIPTION	DESC2
CBL/GS20101	1		732053	MURR	DEV45 335255	PATCH-CABLE, 4 PIN, STRAIGHT M12 MALE TO M8 FEM, 0.6M, PINS 1 & 3 JUMPED M12 PLUG, M12 SOCKET 4PIN CABLE DIAMETER 5.2mm 250V, 4A, 4 CONDUCTORS, BN, WT, BU, BK, PVC JACKET	
PLUG50101 PLUG50102 PLUG50104 PLUG50106	4		732026	MURR	M12-PLUG-PA		
CBL/PRS30209	1		732050	MURR	7000-40021-2240100	PATCH-CABLE, 4 PIN, STRAIGHT M12 MALE TO FEMALE, 1M, GRAY M12 PLUG, M12 SOCKET 4PIN CABLE DIAMETER 5.2mm 250V, 4A, 4 CONDUCTORS, BN, WT, BU, BK, PVC JACKET	
CBL/PRS417	1		743147	MURR	7000-40121-2240060	PATCH-CABLE, 4 PIN, STRAIGHT M12 MALE TO 90-DEG FEMALE, 0.6M, GRAY M12 PLUG, M12 SOCKET 4PIN CABLE DIAMETER 5.2mm 250V, 4A, 4 CONDUCTORS, BN, WT, BU, BK, PVC JACKET	
PJ50100	1		732045	MURR	8000-88510-3980500		CABLE
MTR402 SOL115 SOL222	3		SEE	MECHANICAL	BOM		2.0 AMPS SUPPLIED BY BRENTON SUPPLIED BY BRENTON
LS30214	1		735700	OMRON	Z-15GQ22-B	LIMIT SWITCH - GENERAL PUROSE, PANEL MOUNT SPRING RETURN ROLLER PLUNGER 1 FORM C	
GS20101	1	*1	739147 (SENTRY PLUS)	SICK SENTRY PLUS)	1059505/RE21-SAC		N.O.H.C.
		*1	750576	TONELUCK (SENTRY LP)	VCS-216		
LS30211	1	*1	737310 (SENTRY PLUS)	TELEMECANIQUE	XEP4E1FDA454J03		
		*1	750575 (SENTRY LP)	TONELUCK	VCS-216		
PE30209	1	*1	SEE	LISTED	OPTIONS		
		*1	735702 (STANDARD SENTRY LP)	P&F	205234/RLK31-8-2500-IR/31/15 (PNP)		
		*1	731972 (DARK LOAD)	SICK	1019229/WT34-B410		
PRS417	1		731785	IFM	IG6083	M18 PROX SENSOR, UN-SHIELDED, 0.8-8mm RANGE, 4-20mA ANALOG, 3-WIRE, M12 PROXIMITY SENSOR - INDUCTIVE 15-30VDC BARREL DIAMETER: 18mm, NOMINAL SENSING DISTANCE: 0.8-8mm, UNSHIELDED OUTPUT CONFIG.: 4-20mA ANALOG, MICRO QUICK-DISCONNECT	SUPPLIED BY BRENTON

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DWG TITLE

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MACHINE FIELD AND CARRIAGE BOM

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JOB NO	DRAWN BY
SCALE: NO SCALE	DATE: 3/17/2018

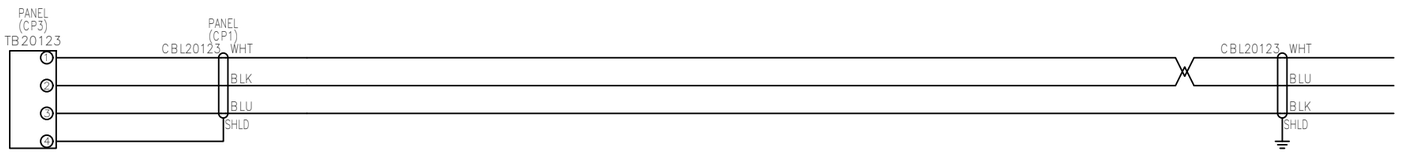
DWG NO

B&R SENTRY

SHEET NO

12 OF 15

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DWG TITLE

FLEX (STANDARD)
B&R SENTRY LP/HP

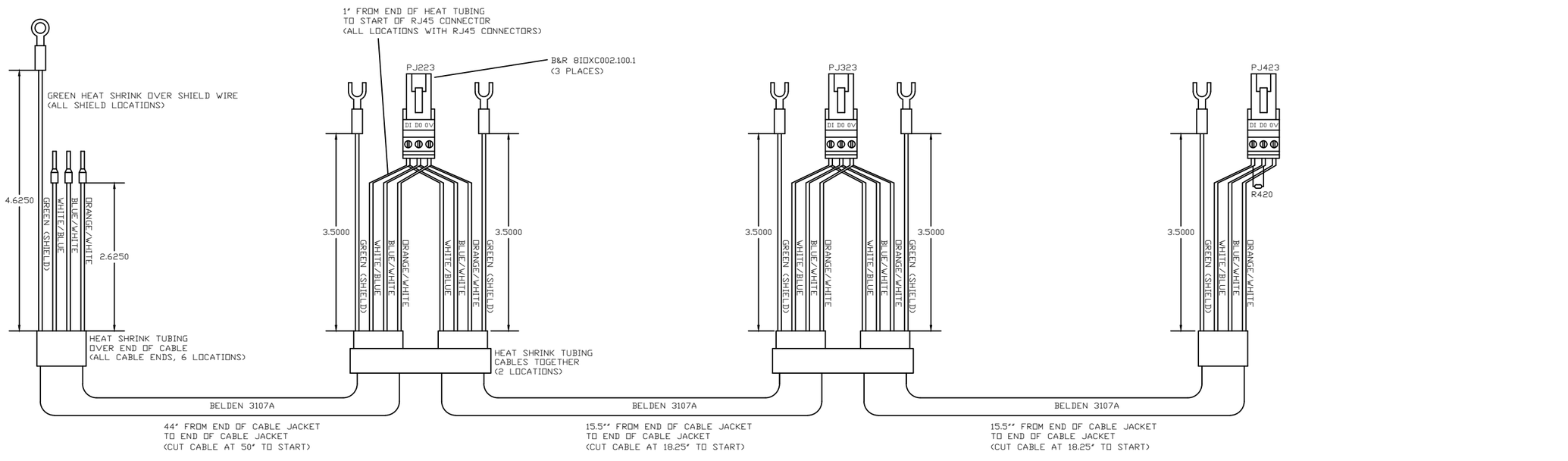
CABLE CBL30502
(743145)

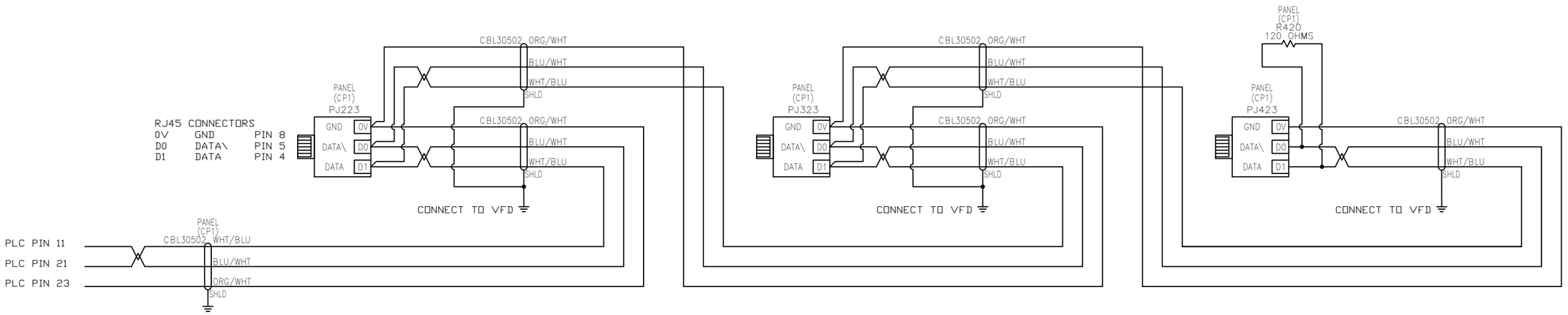
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SCALE: NO SCALE	DATE: 3/17/2018

DWG NO
B&R SENTRY

SHEET NO
13 OF 15

CAD DWG FILE I:\PROJECT_Flex Machine Orders\V6092 NEW SENTRY - Scale, ramp, manual\Electrical\schematics\PG13.dwg





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DWG TITLE

FLEX (STANDARD)
B&R SENTRY LP/HP

CABLE CBL30502
(743145)

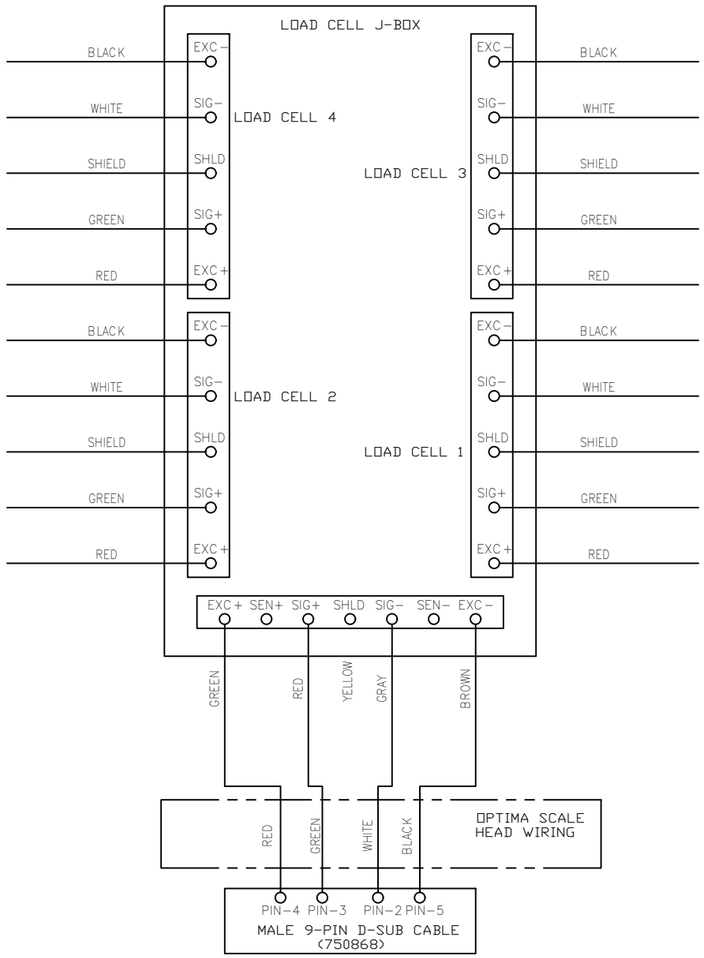
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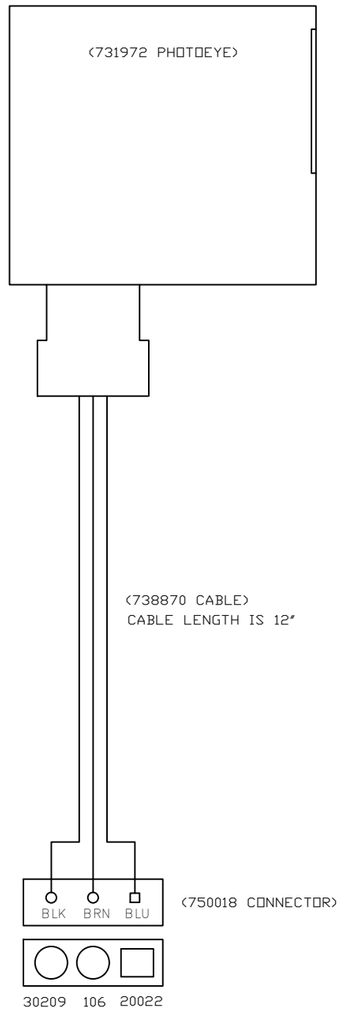
SHEET NO
14 OF 15

CAD DWG FILE I:\PROJECT_Flex Machine Orders\V6092 NEW SENTRY - Scale, ramp, manual\Electrical\schematics\PG14.dwg

TORREY/PROTEK WI-10T SCALE KIT (D0393157



DARK LOAD PHOTOEYE KIT (D0393156)



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DWG TITLE
FLEX (STANDARD) B&R SENTRY LP/HP

OPTIONS PAGE

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SCALE: NO SCALE	DATE: 3/17/2018

DWG NO
B&R SENTRY

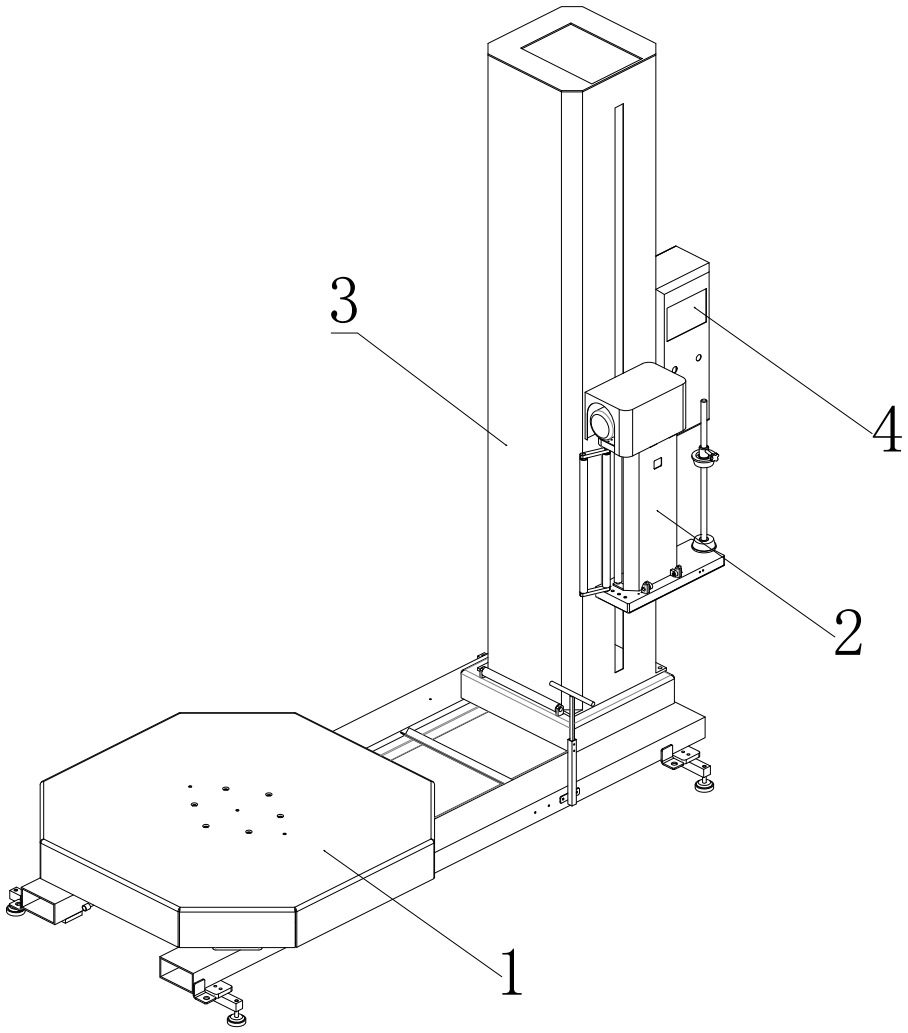
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Mechanical Drawings

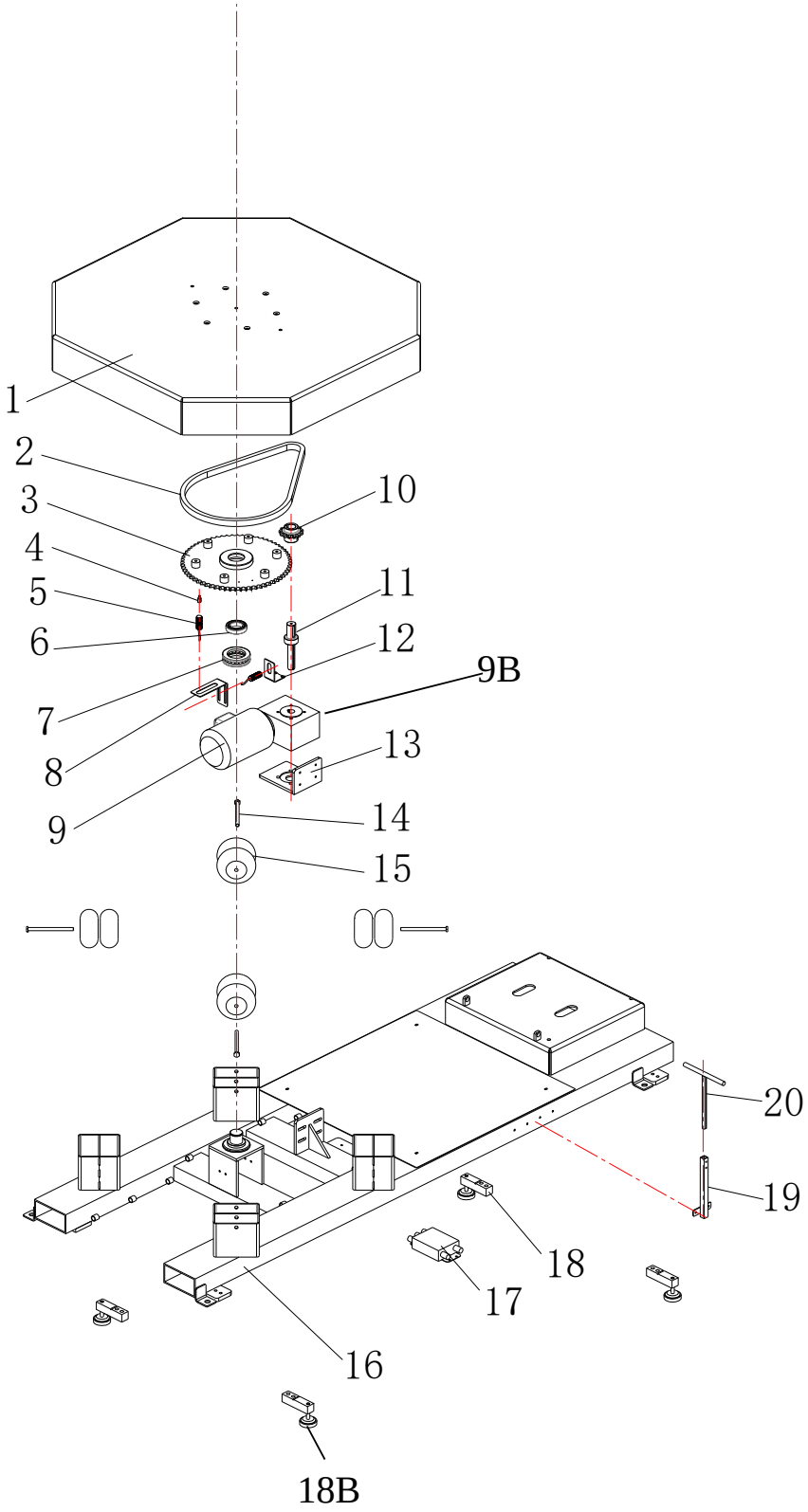
Download the mechanical drawings for your particular model on the Orion Webpage.
https://www.orionpackaging.com/parts-and-service/manuals/

SENTRY HP/ LP PARTS DIAGRAM AND LIST



NO.	Name	Part Number	Qty	Remarks
1A	Sentry HP Base	753831		
1B	Sentry LP Base Assembly	749511		
2	Sentry Carriage Assembly	750497		
3	Sentry Tower Assembly (B&R)	749510		
4	Sentry Control Panel	750029		

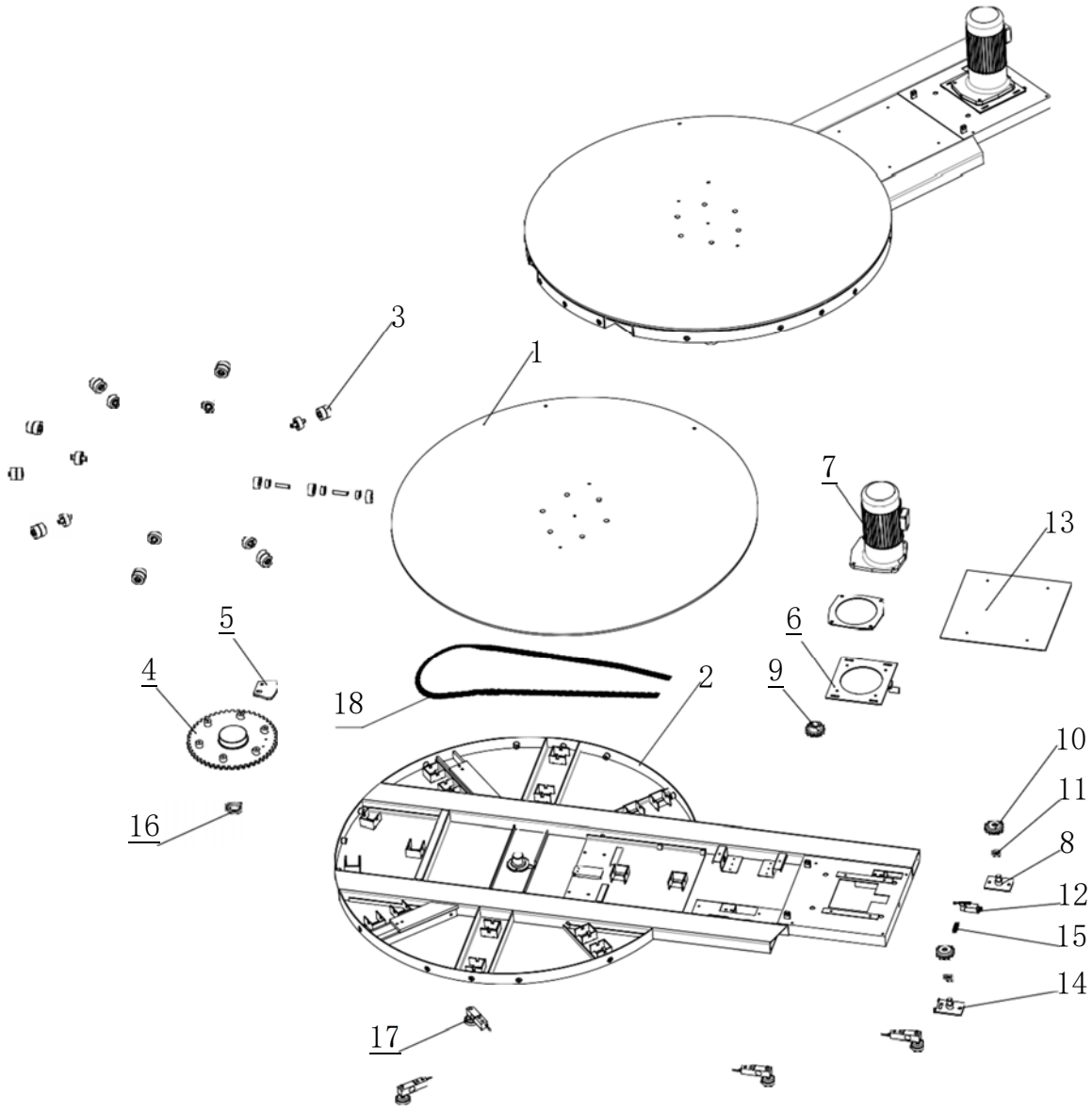
HP Base Unit Assembly



HP Base Parts List

NO.	Name	Part Number	Qty	Remarks
-	Complete HP Base Assembly	753831	1	
1	Turntable	752068	1	
2	TT Chain	752069	1	
3	TT Sprocket	751259	1	
4				
5				
6	Bearing	752070	1	
7	Bearing	752072	1	
8	Proximity Switch Mount	752073	1	
9A	Motor	752074	1	
9B	Gearbox	753841	1	
10	Drive Sprocket	753842	1	
11	Motor Shaft	753843	1	
12	Sensor Bracket	753844	1	
13	Reducer Mount	753845	1	
14	Caster Wheel Shaft	753846	4	
15	Caster Wheel	753847	8	
16	Sentry HP Base	753848	1	
17	Stainless Wire Box	753849	1	
18A	Scale Load Cell	751015	4	
18B	Scale Load Cell Feet	751016	4	
19	Roping Bar Holder	753850	1	
20	Roping Bar	753851	1	
21	Caster Wheel Kit	753852	1	
22	Optional Scale Kit with Feet, Load Cells, and LCD Indicator	751013	1	

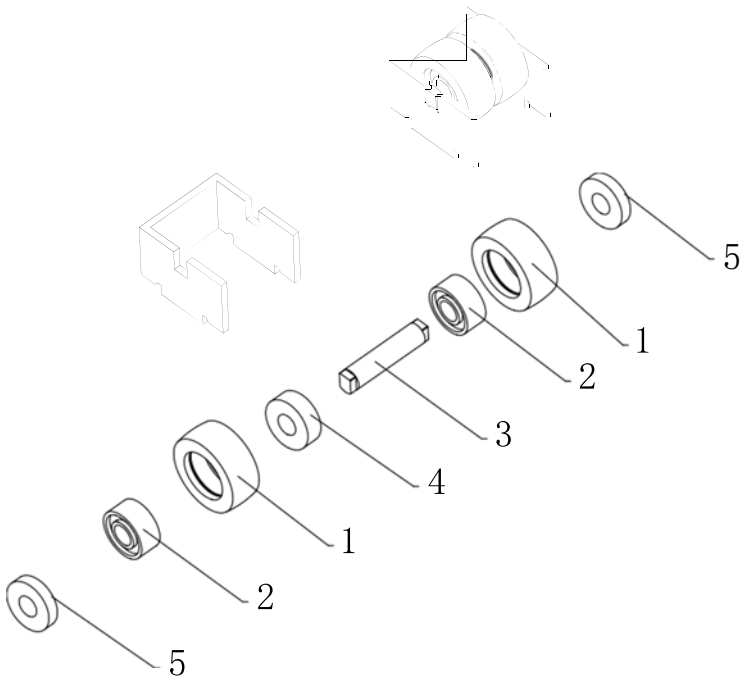
LP Base Unit Assembly



LP Base Parts List

NO.	Name	Part Number	Qty	Remarks
-	Complete LP Base Assembly	749511	1	
1	Turntable	751257	1	
2	Base	751258	1	
3	Roller Assembly	See Roller Assembly	13	
4	Driven Sprocket	751259	1	
5	Travel Switch Touch Block	751260	1	
6	Main Motor Baseplate	751261	1	
7	Turntable Motor	734351	1	750W 1:30
8	Tension Wheel Seat	751262	1	
9	Main Motor Drive Sprocket	751263	1	
10	Tension Chain Sprocket	751264	2	
11	Bearing	013428	2	6205
12	Limit Switch	734379	5	
13	Base Cover Plate	751265	1	
14	Elastic Tension Wheel Seat	751266	1	
15	Tension spring	751267	1	
16	Bearing	747355	1	6009
17	Load Cell Sensor	750865	4	Optional
18	Turntable Chain	0297067	1	#60

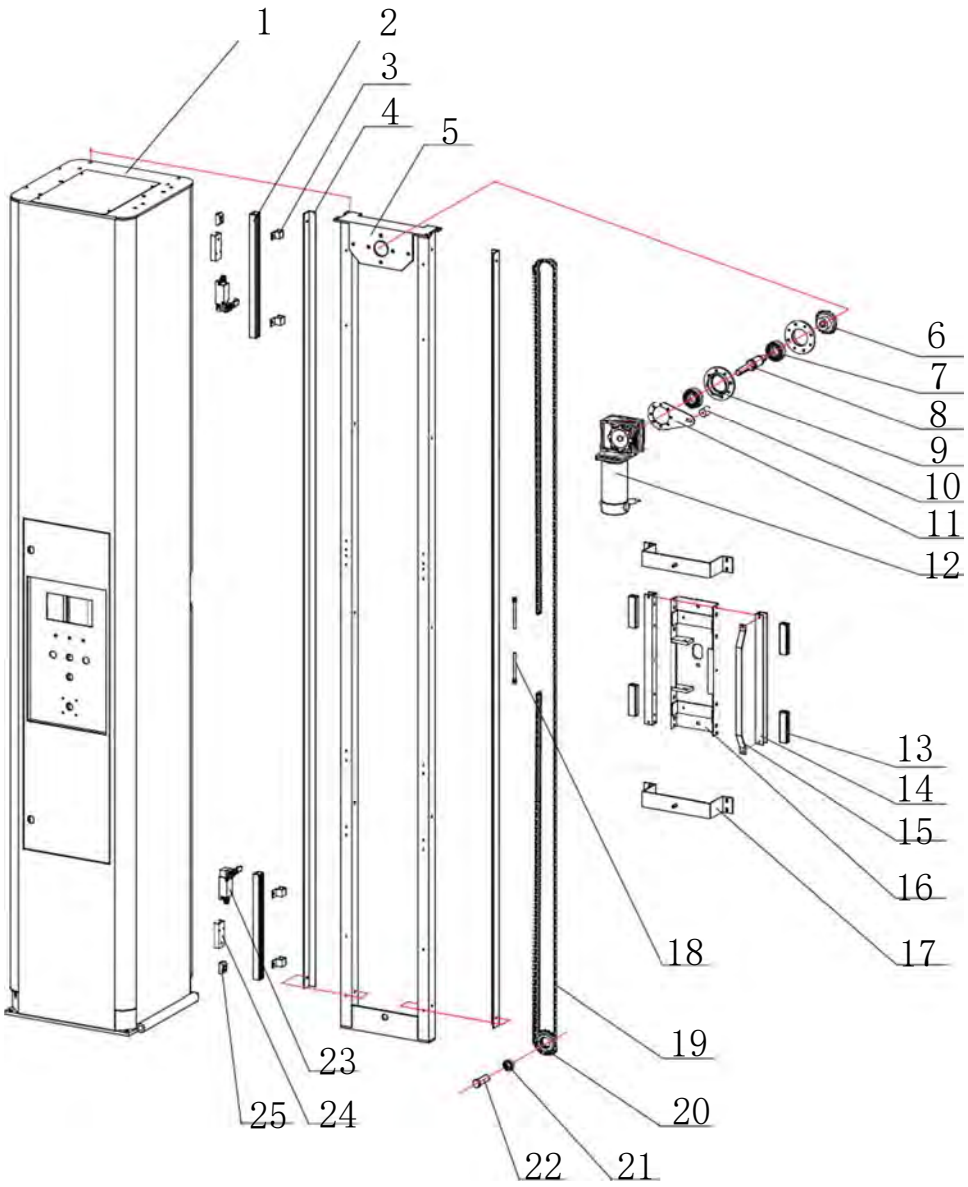
LP Roller Assembly



LP Roller Assembly Parts List

Part Ref	Description	Part Number	Qty	Remarks
1	Roller Body	751229	2	
2	Roller Bearing	712455	2	6202
3	Roller Shaft	751230	1	
4	Inner Spacer Bushing	751231	1	L=16mm
5	Outer Spacer Bushing	751232	2	L=9.5mm
6	Roller Stand	Welded to base Not available for sale	1	

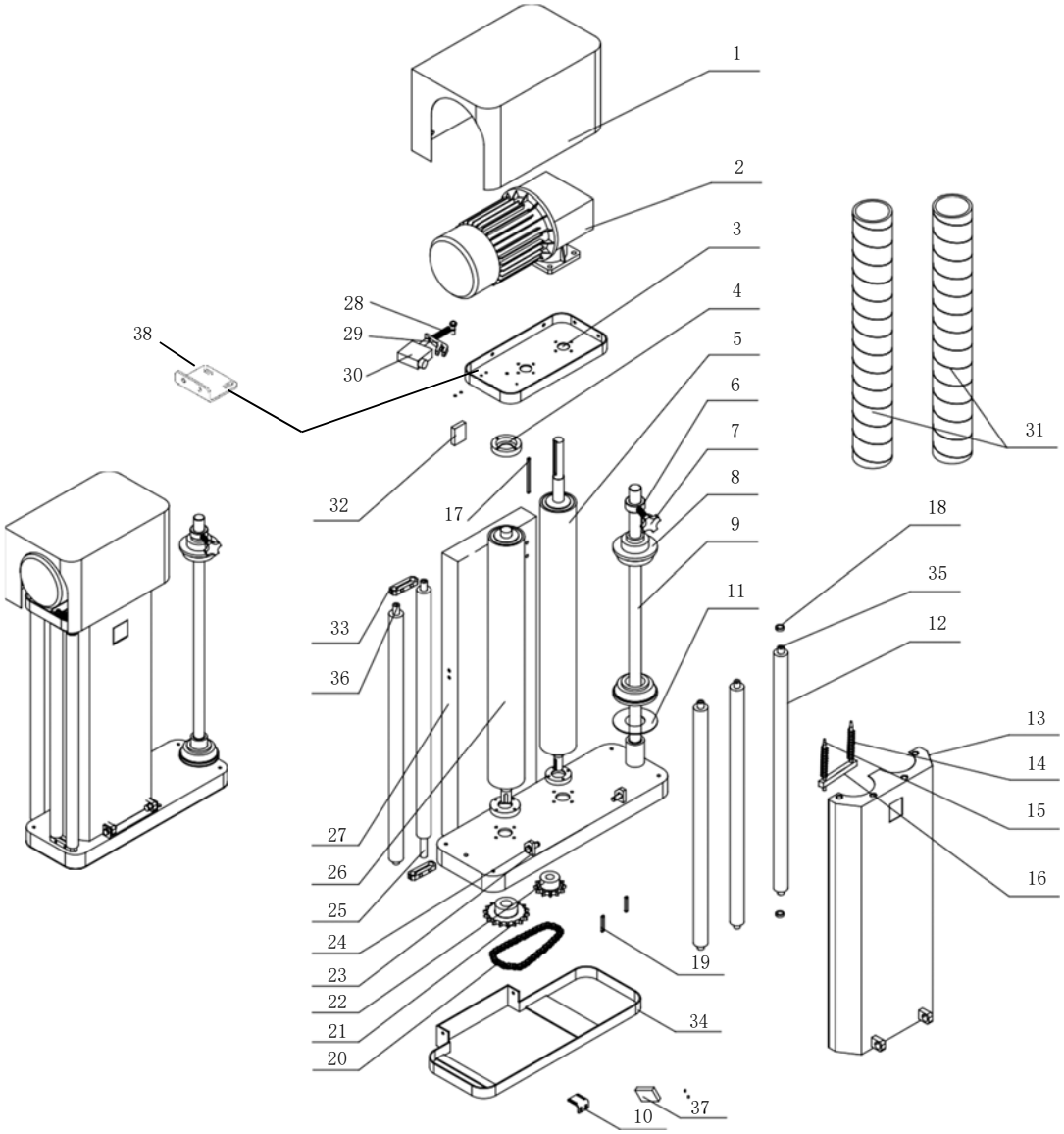
Tower Unit Assembly



Tower Parts List

NO.	Name	Part Number	Qty	Remarks
-	Complete Tower (8’ High) Complete Tower (9’ High)	749510 749991	1 1	
1	Tower (8’ High) Tower (9’ High)	751268 Contact for Details	1 1	
2	Travel Switch Sliding Slot	751269	2	
3	Lifting Wire Slot Support	751270	4	
4	Aluminum Slide	751271	2	
5	Mount Rack Lifting Bracket	751272	1	
6	Chain Driven Sprocket	751273	1	
7	Bearing	013429	2	6206
8	Motor Shaft	751274	1	
9	Bearing Seat	751275	2	
10	Torque Arm Mount	751276	1	
11	Torque Arm	751277	1	
12	Lifting Motor	751278	1	250W
12A	Tower Lift Kit (Motor, Gearbox, Shaft)	1078934	1	
13	Plastic Sliding Block	751279	4	
14	Sliding Block Frame	751280	2	
15	Lifting Touch Block	751281	1	
16	Single Chain Lifting Frame	751282	1	
17	Electrical Mount Plate	751283	2	
18	Adjusting Screw Rod	751284	2	
19	Chain (8’ Tower) Chain (9” Tower)	0364241 0364242	1 1	
20	Chain Idler Sprocket	751285	1	
21	Idler Bearing	741616	1	
22	Chain Idler Shaft	751286	1	
23	Limit switch	751292	2	OMRON
24	Limit Switch Plate	751287	2	
25	Limit Switch Block	751291	2	

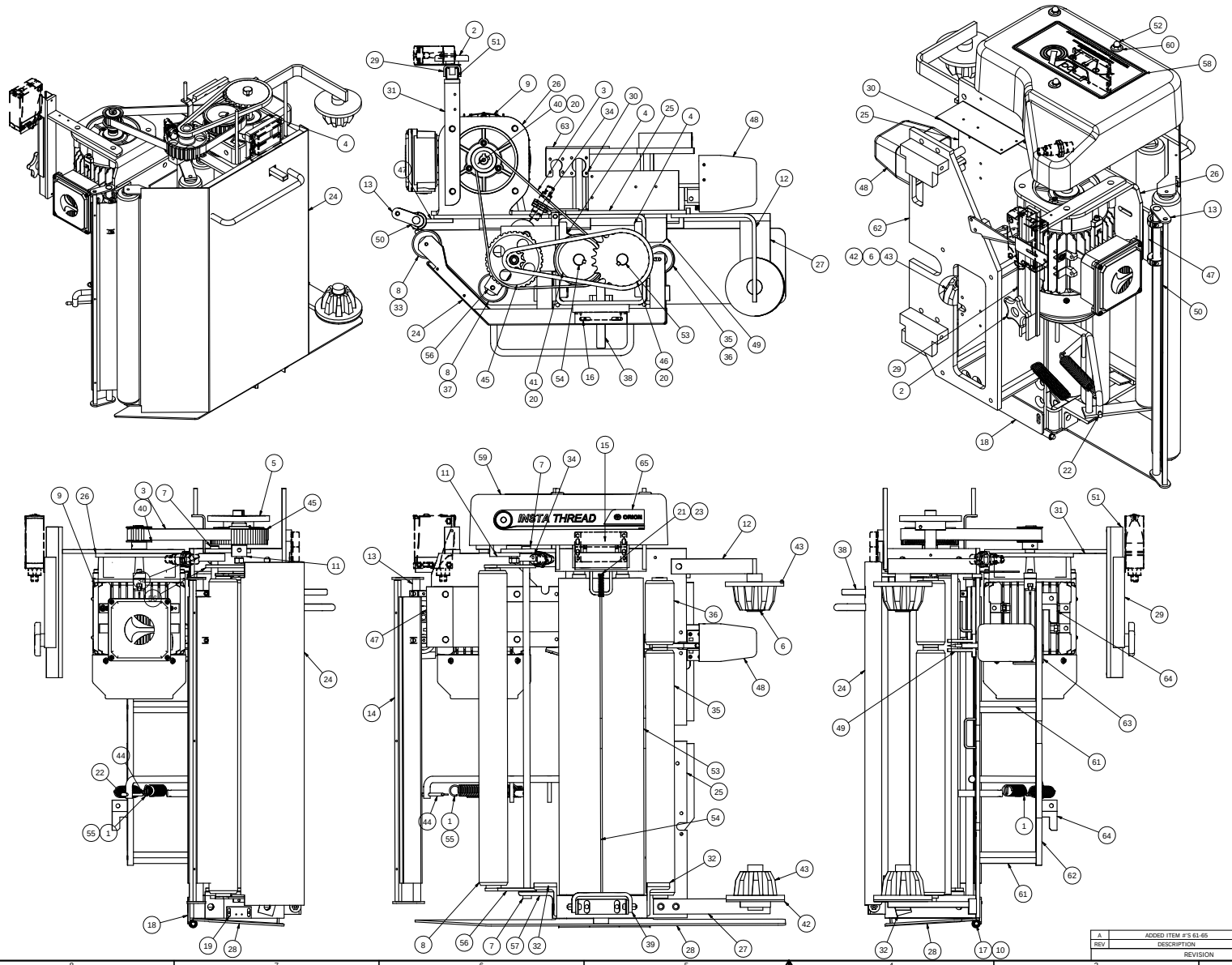
Carriage Assembly



Carriage Parts List

NO.	Name	Part #	Qty	Remarks
-	Complete Sentry Carriage Assembly	750497	1	
1	Pre-stretch Motor Cover	735671	1	
2	Pre-stretch Gearbox & Motor	1011893	1	
2.1	Pre-stretch Motor Only	745000	1	Motor Only(250W)
2.2	Pre-stretch Gearbox Only	735672	1	Gearbox Only
3	Pre-stretch Upper Cover	735673	1	
4	Bearing w/Base	735674	3	
4.1	Bearing Only	741616	3	6904 bearing
5	Pre-stretch Roller	735675	1	
6	Tightening ring	735676	1	
7	Tightening Screw	735677	1	M10
8	Film Locating Plate	735678	2	
9	Film Shaft	735679	1	
10	Micro-Active Switch Frame	735680	1	
11	Locating Plate	735681	1	w/Ext. Ret. Ring
11.1	Ext. Retaining Ring	736040	1	
12	Transition Roller Kit (Rollers, Shaft & Bearings)	735682	5	w/shaft & bearings
13	Cradle Roller Door	735683	1	
14-16	Door Latch Kit (Spring, Pin & Rail)	1011894	--	Item #’s 14,15,16
14	Compressed Spring	735684	2	
15	Locating Pin	735685	2	
16	Cross Bar/Rail	735687	1	
17	Roller Key	753264	1	6mm×6mm×60mm
18	Roller Bearing (Part of item #12)	---------	10	6901
19	Sprocket Key	1070497	2	5mm×5mm×14mm
20	Carriage Chain	735690	1	06B
21	Driven Sprocket	735691	1	
22	Drive Sprocket	735692	1	
23	Lower hinge	735693	1	
24	Hinge pin	735694	2	
25	Swing support shaft	735695	1	
26	Main roller	735696	1	
27	Lifting body	735697	1	
28	Tension spring	735698	1	
29	Micro switch touch block	735699	1	
30	Micro switch	735700	1	OMRON
31	Rough Surface on Main Roller	Contact for Details	2	

32	Top Load Photoelectric switch	735702	1	P+F
33	Left Connecting Block	735703	2	
34	Bottom Protection Frame	735704	1	
35	Transition roller shaft (Part of item #12)	----------	3	
36	Swing shaft (short)	735706	1	
37	Micro switch	738183	1	
38	Dark Photo Eye Mount Bracket	1057079	1	Optional Part



BOTH SPRINGS TO BE SHIPPED WITH MACHINE.
INTERCHANGEABLE IN REFERENCE TO FILM GAGE
AND TENSION NEEDED. (010047 & 745661)

ITEM	QTY	PART NUMBER	DESCRIPTION
65	1	750374	STCK ORION CARRIAGE DESIGN
64	2	1039849	CARRIAGE MOUNT PAD
63	1	1045125	HARNESS BRACKET
62	1	1039823	CARRIAGE BRACKET
61	4	1039828	CARRIAGE STANDOFF
60	3	014502	HDW (CART ITEM) FASTENAL 10034-12836 GROMMET 3/8" ID X 1/4" GRV X 1/2" GRV ID
59	1	414305	HVA CARRIAGE COVER RH (ABS)
58	1	500207	STCK CUSTOM RH THREADING DIAGRAM
57	1	D0245989	HVA BRACKET DANCER ROLLER RH
56	1	D0245988	VGS STFB DANCER ROLLER CRADLE - 20 FLR
55	1	016188	VGS FASTENAL 0166826 EXTENSION SPRING #75 0.75 OD, 0.062 Wire, 4.5 Overall
54	1	420917	RUBBER ROLLER (20" film)
53	1	420916	ROLLER, RUBBER, #1, 20"
51	1	452729	VGS HVF Photo High Channel
50	1	443430	BRUSH ASSY
49	1	443422	INSTA CUT PIN PROTECTOR
48	1	443421	FLX P-ASSY FILM PUNCH DEVICE -INSTA THRD
47	1	443416	ADJUSTED ROLLER HOLDER-1
46	1	435715	FLX SPROCKET 40B24 SPECIAL
45	1	434017	FLX SPRK 40B8/32L075 WELDMENT
44	1	433629	FLX HVF TENSION SCREW
43	2	432323	HVF SPOOL, PLASTIC
42	1	432322	HVF BOTTOM SPOOL WASHER
41	1	431672	HVF GEARBELT PULLEY 26L075
40	1	431477	HVF GEARBELT PULLEY 12L075
39	1	430054	ASSY HINGE CRADLE ROLLER
38	1	430053	VGS HVF LATCH
37	1	423966	IDLER ROLLER SHAFT
36	1	421884	IDLE ROLLER - 1.9 DIA X 4 5/8
35	1	421883	IDLE ROLLER - 1.9 DIA X 16 5/8
34	1	413744	HVF PROX SENSOR CAM
33	3	413249	IDLE ROLLER SHAFT - 21.3125
32	2	409469	HVF CRADLE ROLLER OPENING LOCK
31	1	1039834	CARRIAGE PRX MOUNT
30	1	1039833	CABLE MOUNTING BRACKET
29	1	1039830	PHOTOCELL HOLDER
28	1	1039829	OBSTACLE DETECT BUMPER
27	1	1039827	BOTTOM MANDREL
26	1	1039824	MOTOR BRACKET BASE PLATE (FLR)
25	1	1039815	20 IN CARRIAGE BACKPLATE WELDMENT
24	1	1039808	VGS HVF CRADLE SHELL WELDING FLR-20
23	1	097973	HDW (CI) AB6-6A BLIND RVT PKG
22	1	097921	HDW (CI) 3/16" O.D. Rubber Vacuum Cap
21	1	097810	SPRG (CI) #5 COMP SPRING 0.343 OD, 0.047 WIRE, 1.5 LONG
20	3	096735	HDW (CI) KEY 0.19 X 0.88 LG STEEL NO FINISH
19	1	0342245	OBSTACLE DETECT SWITCH MOUNT
18	1	0328195	OBSTACLE DETECT BUMPER- MOUNT
17	1	0327297	HINGE PIN
16	1	0283560	SICK DOOR MOUNT
15	1	0283554	BACK PLATE SWITCH MOUNT
14	1	0277062	20" INSTA CUT ROTATING BRUSH MOUNT
13	1	0277061	20" INSTA CUT BRUSH MOUNT
12	1	0258573	TOP ROLLER SUPPORT
11	1	D0248103	CARRIAGE CHAIN TENSIONER BRACKET
10	2	015314	VGS RETAINING RING, SELF SEATING, EXT.1/4 R6100-25
9	1	749110	EM TECHTOP GR3-AL-TF-56C-4-B-D-.5
8	3	015153	VGS CONV Roller, #1.775-R79-21.25 Interroll AL 20
7	2	014247	VGS Flange Bushing 7/16 ID X 9/16 OD X 1/2 LG
6	2	013860	RR TRUARC 5105-00 EXT 1" SELF SEATING
5	1	013397	VGS CAN Chain Roller #40 (44 Links+C/L)44 Links + C/L
4	4	011192	VGS BRG IPTCI UCP 204-12Bearing, Pillow Block, 3/4" Bore
3	1	011151	VGS BELT Gear Belt 3/8" Pitch x 3/4" Wide x 28.50
2	1	010092	MSCE (CART ITEM) KNOB W/THREADED BLIND HOLE, 5PTS STAR 3/8-16UNC 3/4 STUD
1	1	010047	SPRG (CI) 0.75 ODx0.08 WIREx4.50L

Parts List

REV	DESCRIPTION	DATE	BY
A	ADDED ITEM #'S 61-65	1/2/19	BAN
	REVISION		

DRAWN rpfau 11/16/2017
CHECKED
APPROVED

Tolerances Unless Noted

Decimal	Angular	Fractional
.00 ± .01	± 1/2'	± 1/32"
.000 ± .005		

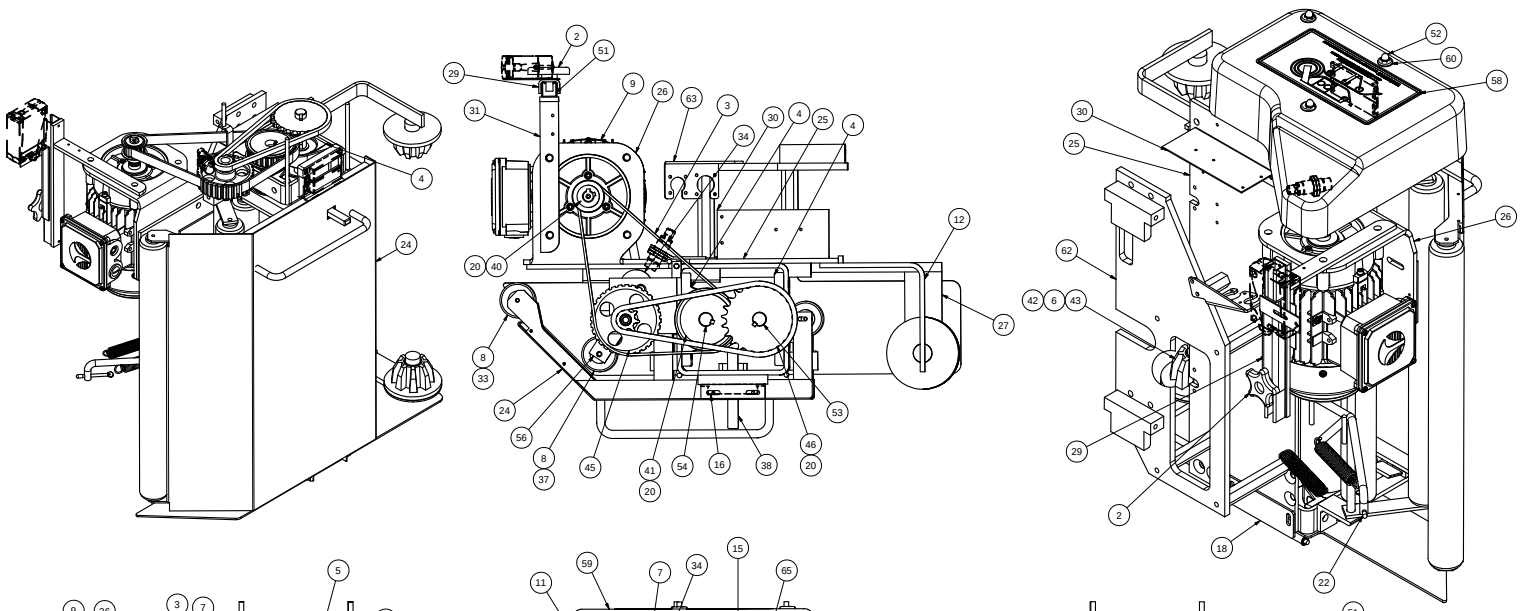
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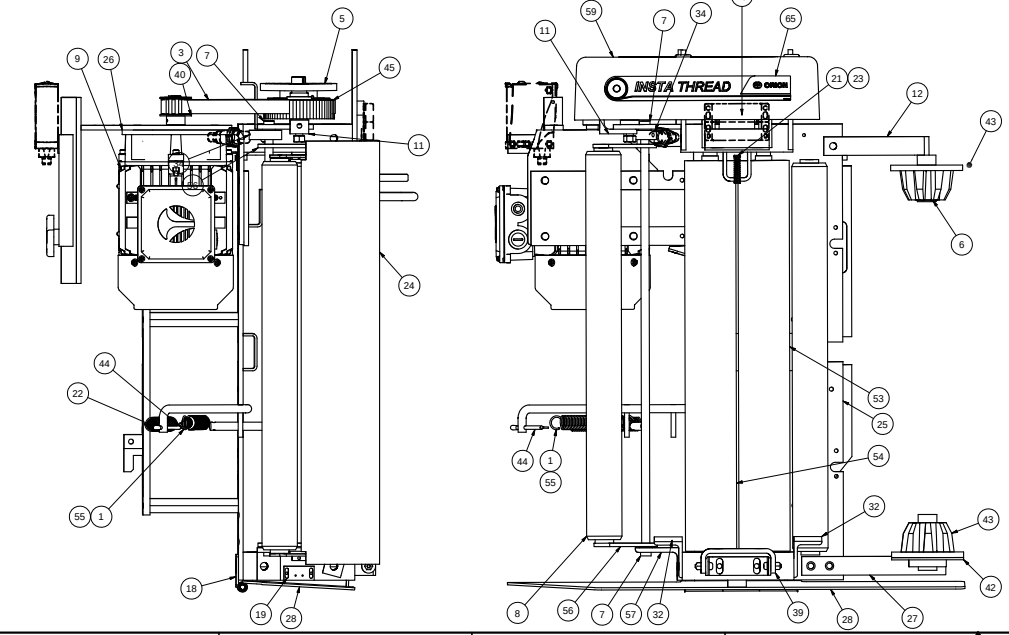
BRENTON ENGINEERING
4750 County Road 13 NE
Alexandria, MN 56308-9014

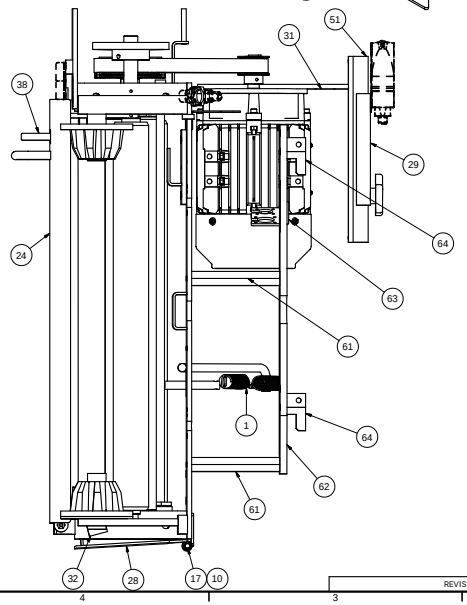
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TITLE: CARRIAGE 20FR-S-IC-260

REVISION	SURFACE AREA	DWG NO	FINISH
		1039822	







BOTH SPRINGS TO BE SHIPPED WITH MACHINE.
INTERCHANGEABLE IN REFERENCE TO FILM GAGE
AND TENSION NEEDED. (010047 & 745661)

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Parts List

DRAWN: wlang 11/23/2019
CHECKED:
APPROVED:

Tolerances Unless Noted

Decimal	Angular	Fractional
.00 ± .01 .000 ± .005	± 1/2°	± 1/32"

REMOVE ALL BURRS AND SHARP EDGES

REVISION

BRENTON ENGINEERING
4750 County Road 13 NE
Alexandria, MN 56308-9014

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TITLE: CARRIAGE 20FR-S-260

REVISION	SURFACE AREA	DWG NO	FINISH
		1085478	